Polarizations and Descent in a Finite Family of CM Sixfolds

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Abstract

Starting from the primitive six-dimensional factor of the Jacobian of $y^2 = x^{21} - 1$, we classify a finite family of integral quotients and minimum polarizations. The source and its order-27 quotients give 189 underlying complex isomorphism classes and 2969 minimum polarized classes. Of the 188 quotient classes, 80 are principalizable and 108 are not; we determine their minimum forms and distinguish target counts from kernel counts. We compute the Galois action and fields of moduli, then prove a descent obstruction and construct a common field of definition of degree 24. Minimum absolute definition degrees are sharp for 168 quotient classes, with 20 cases left as lower bounds. A restricted ten-class Jacobian classification is stated under its explicit enclosure and execution assumptions; 156 smaller-order principal Jacobian questions remain open.

Companion references. Result numbers prefixed I, II, H and S refer respectively to the packet paper, quotient paper, supporting Hodge note and computational supplement. Each document has its own numbering and bibliography.

Contents

1 Introduction . . . 2
2 The source factor and its polarization . . . 3
3 Minimum isogenies and the minimum source polarization . . . 6
4 Principal and underlying quotient classifications . . . 8
5 Minimum forms on nonprincipalizable quotients . . . 12
6 Galois action fields and orbit classification . . . 14
7 Real models and actual descent . . . 19
8 A common field of definition . . . 20
9 Minimum degrees and the remaining cases . . . 22
10 Weil structures and maps from obstructed quotients . . . 23
11 A structural two-prime comparison . . . 25
12 The restricted principal census and Jacobian question . . . 26
13 The smaller-order Jacobian cases . . . 30
14 Questions left by the classification . . . 31

1 Introduction

This paper asks an integral question within one complex CM isogeny class: which quotients of a fixed sixfold admit principal polarizations, and over which fields can the quotients and their polarizations be defined? A rational Hodge-algebraicity theorem does not answer either question. The starting variety is defined from a curve, independently of any Hilbert-modular packet.

Let H be the smooth projective model of $y^2 = x^{21} - 1$. Its primitive Φ_{21} factor $A \subset J(H)$ is six-dimensional and has full endomorphism ring $\mathcal{O}_M$, where $M = \mathbb{Q}(\zeta_{21})$. We keep this actual embedded factor distinct from isogenous principal models. Its restricted Jacobian polarization has degree 35721, whereas its minimum polarization has degree 729 and type $(1, 1, 1, 3, 3, 3)$.

1.1 The finite family and the equivalence relations

Fix the minimum polarization $\lambda_{\min}$ and write its kernel as $A[\pi] \simeq \mathbb{F}_3^6$, with the notation made explicit below. The family consists of A and all quotients A/H_0 for subgroups $H_0 \subset A[\pi]$ of order 27. All isomorphisms preserve the origin. An underlying class forgets the polarization; a polarized class retains it but forgets CM and RM markings. A kernel is a subgroup of the fixed source, not a target isomorphism class. These distinctions account for the different counts in the following summary.

Object being counted	Number
Order-27 subgroups of the fixed source kernel	33880
Underlying quotient classes (excluding A)	188
Underlying classes including A	189
Principalizable underlying quotients	80
Principal polarized quotient classes	160
Nonprincipalizable underlying quotients	108
Minimum polarized classes including A	2969

The last number is $1 + 160 + 108 \cdot 26$. The source contributes its unique minimum polarization, and each obstructed quotient contributes 26. Fixing the pullback form gives 1120 principal-target kernels; allowing all minimum pullback forms gives 14224. Neither number counts all 33880 order-27 subgroups as principal-target kernels.

1.2 Main results and proof route

Theorem 4.1 classifies the principal targets; Theorem 4.3 forgets the polarizations and identifies the obstruction; Theorem 5.1 determines the minimum forms on the remaining 108 varieties. Their proofs pass from integral lattices to finite residue subspaces and distinguish the relevant unit actions before taking orbits.

The Galois action then determines fields of moduli, but a fixed isomorphism class need not descend. Put

$$\mathcal{E} = M(\sqrt{(3 + \sqrt{21})/2}).$$

Theorem 8.1 proves that for number fields $M \subseteq F \subset \overline{\mathbb{Q}}$, any one of the 188 quotient classes has an F-model exactly when F contains $\mathcal{E}$. A single twist makes all subgroup kernels and quotient maps available over $\mathcal{E}$. Thus the common field-of-moduli computation and the actual defining-field construction are distinct stages of the argument. Theorem 9.1 gives sharp absolute degrees for 168 quotient classes and lower bounds for the other 20.

The final geometric section treats Jacobians. It distinguishes the restricted ten-class principal census with R-containment from the 156 principal classes with smaller endomorphism order S. The former has exactly one canonical Jacobian under the stated enclosure and execution contract. The latter 156 statuses remain open.

1.3 Arithmetic inputs and relation to other work

Write $z = \zeta_{42} = -\zeta_{21}^{11}$, $t = z + z^{-1}$, $L = M^+$ and

$$f(T) = T^6 + T^5 - 6T^4 - 6T^3 + 8T^2 + 8T + 1, \qquad \mathfrak{P} = (1 - z^{14})\mathcal{O}_M, \quad R = \mathcal{O}_L + \mathfrak{P}, \quad S = \mathbb{Z} + \mathfrak{P}.$$

Here $\mathfrak{P}$ is the prime over 3; its residue field is $\mathbb{F}_{729}$ and $R/\mathfrak{P} = \mathbb{F}_{27}$. The letter π below is $1 - z^{14}$, a generator of $\mathfrak{P}$ in M. It is *not* the norm-43 generator of Paper I. We use $K = \mathbb{Q}(\sqrt{-3})$ for the Weil action. Write $E_L = \mathcal{O}_L^\times$ and let E_L^+ denote its subgroup of totally positive units.

The imported arithmetic inputs are the ideal classification and marked polarization classification of Paper I, Propositions I.7.9, I.7.10 and I.7.11, and the unit-norm theorem in the supplement. They are established without using this quotient family. The notations B_0, B_1, C_0, C_1 mean the two principal-polarization classes on each of the two R-ideal models in the B/C CM-type orbit; Paper I identifies them with its principal curve fibers. The quotient construction here needs their arithmetic classification, not that curve placement. The type representatives are $\Phi_B = \{1, 5, 23, 25, 29, 31\}$ and $\Phi_C = \{1, 5, 11, 13, 17, 19\}$; $\Phi_0 = \Phi_C$.

Goodson and Saller discuss the rational CM-type/Fermat comparison [3, 15]. The general CM Hodge assertion in [13], if valid, concerns algebraicity of rational classes; it does not itself provide the integral census or descent data here. The existence of individual models must also be compared with classical CM descent, including Milne's correction and addendum [6, 7]. Our explicit simultaneous twist, field obstruction and finite tables are distinguished from that existence question. These comparisons are not a certification of historical priority.

2 The source factor and its polarization

Let $M = \mathbf{Q}(\zeta_{21}) = \mathbf{Q}(\zeta_{42})$, with $\zeta_{42} = -\zeta_{21}^{11}$. The following construction separates a rational CM-type comparison from the principal polarizations classified in Section 12. The order-and-type census used here is proved in Theorem 12.1; its proof uses only the imported order arithmetic, not the quotient or descent conclusions.

Proposition 2.1. *Let H be the smooth projective curve with affine equation $y^2 = x^{21} - 1$. Put $\sigma(x, y) = (\zeta_{21}x, y)$ and let $\sigma_J([p - \infty]) = [\sigma(p) - \infty]$ on $J(H)$. The connected image A of a denominator-cleared primitive Φ_{21} idempotent is a simple abelian sixfold with $\mathrm{End}(A) = \mathcal{O}_M$. Its rational CM type belongs to the B/C orbit, but A admits no principal polarization. In particular, the polarization restricted from $J(H)$ is nonprincipal.*

Proof. The double cover of the x-line has twenty-two simple branch points, including infinity, so H has genus ten. A basis of holomorphic forms is $x^{i-1}dx/y$, $1 \le i \le 10$. At infinity the orders of x, y, dx are $-2, -21, -3$, respectively, giving order $20 - 2i \ge 0$ for these forms; at a finite branch point dx/y is regular in the parameter y. The eigenvalue of σ^* on the ith form is ζ_{21}^i. The primitive characters are exactly

$$T = \{1, 2, 4, 5, 8, 10\}.$$

Their complex conjugates complete the twelve primitive characters in rational cohomology. Abel–Jacobi equivariance identifies these with the cotangent eigenvalues of the stated σ_J; the derivative

on the tangent space and its dual on cotangent forms have the same eigenvalues. This convention is essential: functorial pullback on $\mathrm{Pic}^0(H)$ would give σ_J^{-1}.

The Chinese remainder decomposition of $\mathbf{Q}[t]/(t^{21}-1)$ gives a rational idempotent e for Φ_{21}, acting by $t=\sigma_J$. Choose an integer $N>0$ such that Ne is an integral endomorphism. The image of this abelian-variety homomorphism is a connected abelian subvariety; its tangent space is the image of the derivative. Thus $A=\mathrm{im}(Ne)$ has dimension six. Its image does not depend on N, since multiplication by a nonzero integer is surjective on an abelian variety. The tangent CM type for $\zeta_{21}\mapsto\sigma_J|_A$ is T. Under $\zeta_{42}=-\zeta_{21}^{11}$ its exponents become $\{1,5,23,25,29,31\}$, the actual B type.

Both σ_J and its inverse preserve A, since they commute with e. Their restriction is an integral automorphism with minimal polynomial Φ_{21}, so $\mathcal{O}_M=\mathbf{Z}[\zeta_{21}]\subseteq\mathrm{End}(A)$. The primitive-type endomorphism calculation in Theorem 12.1 gives simplicity and $\mathrm{End}^0(A)=M$ without requiring a principal polarization. Maximality of $\mathcal{O}_M$ then gives equality integrally. If A admitted a principal polarization, it would occur in Theorem 12.1's complete census: its endomorphisms contain the specified order R, and the Rosati involution on the CM field is complex conjugation. But the B/C type orbit has no principal object with full endomorphism order $\mathcal{O}_M$. This contradiction proves the assertion. Restriction of the ample Jacobian polarization is a polarization on A, hence is nonprincipal. □

There is also a direct Fermat domination. For $c^{21}=4$, the formula

$$(u,v)\longmapsto(X,Y)=(cuv,\ \mathrm{i}(u^{21}-v^{21}))$$

on $u^{21}+v^{21}=1$ satisfies $Y^2=X^{21}-1$. The nonconstant map extends to the smooth projective models. The identity $f_*f^*=[\deg f]$ on Jacobians exhibits $J(H)$, and therefore A, as isogeny factors of the Fermat Jacobian. This does not identify A with any of the four principally polarized B/C objects: their full endomorphism order is R, whereas that of A is $\mathcal{O}_M$. Proposition 2.2 below computes the actual restricted polarization. Proposition 3.1 determines the minimum unpolarized isogenies to B/C. A polarized identification with a particular prior-art factor remains a separate question. Their genus-six Jacobian status is instead addressed by Theorem 12.4, under its stated computational contract.

Proposition 2.2 (The induced polarization on the Fermat factor)**.** *For the actual embedded primitive factor $A\subset J(H)$ above, the restriction of the canonical principal polarization has type $(1,1,1,3,3,21)$ and degree 35721. Put $H_d: y^2=x^d-1$ for $d=3,7$, taking smooth projective models, and let $f_d: H\to H_d$ be $(x,y)\mapsto(x^{21/d},y)$. The addition map*

$$A\times J(H_3)\times J(H_7)\longrightarrow J(H),\qquad (a,u,v)\longmapsto a+f_3^*u+f_7^*v$$

is an isogeny of degree 35721.

Proof. The lower curves have genera one and three. Each f_d has prime degree $q=21/d$ and is ramified: at $(0,\pm\mathrm{i})$ the local parameter x maps to x^q. The norm identity gives $f_{d,*}f_d^*=[q]$. If $\ker f_d^*$ contained a nonzero line bundle L, it would have exact order q. Its connected cyclic étale cover becomes trivial after pullback to H, so f_d would factor through that cover. The remaining map would have degree one, contradicting ramification. Thus f_d^* is an embedding. The canonical polarization adjunction, equivalently the degree formula for the integral cup pairing, gives induced form $q\lambda_{H_d}$ on its image Z_d.

Let $T=\sigma_J$ in the covariant convention above, with compatible roots $\zeta_d=\zeta_{21}^{21/d}$. Equivariance implies that $\Phi_d(T)$ annihilates Z_d: the sum of the order-d rotations factors through the Jacobian

of the quotient y-line, which is zero. The integral identity

$$\Phi_7(t) - (t + t^4)\Phi_3(t) = 1$$

therefore proves $Z_3 \cap Z_7 = 0$. Since T preserves the canonical alternating form, eigenvectors pair to zero unless their eigenvalues are inverse. Primitive third and seventh characters are orthogonal. Thus $Z = Z_3 + Z_7$ is the embedded product, with restricted form $7\lambda_{H_3} \oplus 3\lambda_{H_7}$. Its polarization kernel is

$$(\mathbb{Z}/7)^2 \oplus (\mathbb{Z}/3)^6 \simeq (\mathbb{Z}/3)^4 \oplus (\mathbb{Z}/21)^2,$$

so its type is $(1, 3, 3, 21)$.

The remaining rational homology characters are exactly the primitive twenty-first characters. They identify the actual A with the connected orthogonal complement of Z. To keep the integral step explicit, let $\Lambda = H_1(J(H), \mathbb{Z})$ and let M, N be its saturated sublattices in the rational spaces of Z, A. The ambient form is unimodular. Projection of Λ onto $M_{\mathbb{Q}}$ maps onto the dual lattice $M^{\#}$: every integral functional on the primitive sublattice M extends to Λ, and unimodularity realizes it by pairing with a lattice element. The projection kernel is N, whence

$$\Lambda/(M \oplus N) \simeq M^{\#}/M \simeq N^{\#}/N.$$

The two restrictions have the same nonunit elementary divisors. The six-dimensional complement therefore adds two unit entries to the type of Z, giving $(1, 1, 1, 3, 3, 21)$. The addition kernel is the displayed quotient, of order $7^2 3^6 = 35721$, which is also the degree of the restricted polarization on A. □

The degrees here are degrees of the specified polarization and addition map; the product of the six polarization elementary divisors is 189. The minimum unpolarized B/C isogeny degree is instead 27 by Proposition 3.1. This structural argument uses neither the theta computation nor its analytic-library contract.

Corollary 2.3 (An integral primitive projector)**.** *With $T = \sigma_J$, the endomorphism*

$$\mathcal{N} = 21 - 3\sum_{j=0}^{6} T^{3j} - 7(1 + T^7 + T^{14})$$

has image A, is symmetric for the Rosati involution of $J(H)$, and satisfies $\mathcal{N}^2 = 21\mathcal{N}$. For the primitive projector $e = \mathcal{N}/21$, the least positive integer n such that $ne \in \mathrm{End}(J(H))$ is 21.

Proof. The deck-group identities for the actual covers give

$$f_3^* f_{3,*} = \sum_{j=0}^{6} T^{3j}, \qquad f_7^* f_{7,*} = 1 + T^7 + T^{14}.$$

The first sum acts by 7 on the order-three characters and by zero on the others; the second acts by 3 on the order-seven characters and by zero on the others. Thus $\mathcal{N}/21$ is precisely the orthogonal projector onto the primitive part. Multiplication by 21 is surjective on A, proving the image assertion. Rosati sends T to T^{-1} and preserves both subgroup sums. This proves symmetry; the projector identity gives $\mathcal{N}^2 = 21\mathcal{N}$. Equivalently, the polynomial residual is $42(1 + T + \cdots + T^{20})$, which vanishes on $J(H)$ because the full rotation quotient is the projective y-line.

Write $L_A = H_1(A, \mathbb{Z})$ and $\Lambda = H_1(J(H), \mathbb{Z})$. The saturated-lattice projection argument in Proposition 2.2 gives $e\Lambda = L_A^{\#}$. Consequently ne preserves Λ exactly when $nL_A^{\#} \subseteq L_A$. This is

equivalent to n annihilating $L_A^{\#}/L_A$, whose exponent is 21. Lattice preservation gives a holomorphic endomorphism of the complex torus, hence an algebraic endomorphism of this abelian variety. Therefore 21 is necessary and sufficient. □

This is minimality of the denominator of this specified projector; it is not a minimal isogeny statement for the principal B/C objects.

3 Minimum isogenies and the minimum source polarization

Proposition 3.1 (Minimum isogenies from the Fermat factor)**.** *Let A be the actual primitive factor of $J(H)$ above, and let B, C denote the two underlying sixfolds of the four principal objects B_0, B_1, C_0, C_1. Among origin-preserving complex unpolarized isogenies, the minimum degree in each of the directions $A \to B$, $B \to A$, $A \to C$, $C \to A$ is 27. Every minimum kernel is isomorphic to $(\mathbb{Z}/3)^3$.*

For the fixed source A, the minimum kernels giving a quotient isomorphic to B or C are exactly the 28 one-dimensional $\mathbb{F}_{27}$-subspaces of $A[\pi] \simeq \mathcal{O}_M/\mathfrak{P} = \mathbb{F}_{729}$, where $\pi = 1 - z^{14}$: fourteen give each underlying target. For each fixed source B or C, there is exactly one minimum kernel giving a quotient isomorphic to A.

Proof. Put $O = \mathcal{O}_M$. Since $\mathrm{End}(A) = O$ and $\mathrm{Pic}(O) = 1$, its actual rank-one CM lattice can be scaled to O. At a common CM type, the target lattice can be scaled to a line lift I with

$$\mathfrak{P} \subset I \subset O, \qquad IO = O, \qquad [O : I] = 27.$$

These are isomorphisms of complex tori. The needed field re-marking preserves $O, R, \mathfrak{P}$ and the two classes of $\mathrm{Pic}(R) \simeq \mathbb{Z}/2$. Thus the actual B, C classes remain $[\mathfrak{a}], [R]$, respectively, by Propositions I.7.9, I.7.10 and I.7.12.

Reduction modulo $\mathfrak{P}$ proves $(I : O) = \mathfrak{P}$: a nonzero residue scalar cannot send the whole field $\mathbb{F}_{729}$ into a proper $\mathbb{F}_{27}$-line. The equality $IO = O$ gives $(O : I) = O$. Multiplication by π and inclusion supply isogenies in the two directions. Every other unmarked isogeny is scalar in these common coordinates, since composing with a rational inverse of either one lands in the full endomorphism algebra M. Therefore their degrees are exhausted by

$$[I : cO] = \frac{|N_{M/\mathbb{Q}}(c)|}{27} \quad (c \in \mathfrak{P} \setminus \{0\}), \qquad [O : cI] = 27|N_{M/\mathbb{Q}}(c)| \quad (c \in O \setminus \{0\}).$$

The first minimum is $729/27 = 27$, attained exactly for $c = \pi u$, $u \in O^\times$; the second is 27, attained exactly for $c \in O^\times$.

The forward kernel is $\pi^{-1}u^{-1}I/O$. Multiplication by π identifies it with the line $u_{\mathrm{res}}^{-1}(I/\mathfrak{P})$ in $O/\mathfrak{P}$, where u_{res} is the residue of u. The backward kernel is always O/I. Both groups are one-dimensional over $\mathbb{F}_{27}$. The 28 projective lines form a torsor for $\mathbb{F}_{729}^\times/\mathbb{F}_{27}^\times$; the full global-unit image has order 14 by Proposition I.7.9. Its two orbits give precisely the two ideal classes. Conversely every line lift I' yields a kernel $\pi^{-1}I'/O$, whose quotient is scalar-isomorphic to the torus of lattice I'. This proves exhaustion and the split $14 + 14$. The backward formula proves uniqueness for each fixed source. Changing the scalar normalization permutes the same lines. □

These are counts of subgroups and minimum degrees of unpolarized maps. They impose no compatibility with the specified polarizations and do not specify a field of definition or a correspondence of curves.

Proposition 3.2 (Minimum polarization and principal quotients)**.** *The minimum degree of a polarization on the actual primitive factor $A \subset J(H)$ is 729. There is exactly one such polarization up to origin-preserving automorphism of A; its type is $(1,1,1,3,3,3)$ and its kernel is $A[\pi]$.*

Fix a representative $\lambda_{\min}$. Each of the 28 $\mathbb{F}_{27}$-lines in $A[\pi]$ is maximal isotropic and gives a principally polarized quotient with pullback $\lambda_{\min}$. These quotients have exactly four polarized isomorphism classes, each realized by seven distinct kernels: they are B_0, B_1, C_0, C_1.

Proof. Normalize the actual CM lattice of A to $O = \mathcal{O}_M$ as in Proposition 3.1. The rotation preserves the restricted Jacobian polarization and generates M, so its Rosati involution is complex conjugation. Any other polarization differs from it by an element of the commutative algebra $\mathrm{End}^0(A) = M$, and therefore induces the same Rosati involution. Thus the trace-form classification applies to every polarization, without an additional compatibility assumption.

Write $\kappa_O = 1/(f'(t)(z-\bar{z}))$, so $O^\vee = \kappa_O O$. Integral alternating forms have parameters $\xi = \kappa_O a$ with $a \in \mathcal{O}_L$. At the actual B type, positivity forces the sign vector of a to equal that of $\mathrm{Im}\,\kappa_O$, which has odd negative parity by Proposition I.7.11. Hence $N_{L/\mathbb{Q}}(a) < 0$, and the degree is $N_{L/\mathbb{Q}}(a)^2$ because the dual lattice is $a^{-1}O$.

We claim that no real integer has negative norm of absolute value less than 27. Since $\mathcal{O}_L = \mathbb{Z}[t]$, the following reductions of its defining polynomial determine the possible prime ideal norms below 27:

p	$f \bmod p$
2	irreducible of degree 6
3	$(t^3 - t^2 + t + 1)^2$
5	$(t^3 - t - 2)(t^3 + t^2 + 2)$
7	$(t-1)^6$.

The displayed cubics are irreducible by absence of roots. Modulo 2, division by all fourteen monic polynomials of degrees at most three verifies irreducibility. For $p = 11, 13, 17, 19, 23$, direct evaluation gives no root, so the prime ideal norms are at least $p^2 > 27$. This exhausts all rational primes below 27. The finite reductions, root checks and norms here are reproduced exactly by `certificates/fermat_minimum_polarization_norm_v154.py`; that certificate supplies arithmetic inputs to the geometric argument. The only prime ideal of norm below 27 is consequently the one of norm 7, generated by $t-1$, whose field norm is $f(1) = 7$. An ideal of norm below 27 is either this ideal or the unit ideal. All real units have norm $+1$ by Lemma S.11.2, so all their generators have positive norm, proving the claim.

Every polarization therefore has degree at least 729. A minimum degree-27 map of Proposition 3.1 to principally polarized B pulls back a polarization of degree 729. Every minimum parameter a generates the unique real prime above 3, which has ideal norm 27 and is generated by β with $N_{L/\mathbb{Q}}(\beta) = -27$. Relative to one positive minimum parameter, all others differ by E_L^+. Theorem S.11.3 gives $N_{M/L}(O^\times) = E_L^+$, proving uniqueness up to automorphism. Since $\beta O = \mathfrak{P} = \pi O$, the kernel is $\pi^{-1}O/O \simeq (\mathbb{Z}/3)^6$, giving the stated type.

The discriminant pairing on this kernel is alternating over $\mathbb{F}_3$ and self-adjoint for the real residue subfield $k = \mathbb{F}_{27}$. Thus for $a \in k$, $b(x, ax) = b(ax, x) = -b(x, ax)$, whence $b(x, ax) = 0$. Every k-line is therefore isotropic and, having order 27, maximal isotropic. Its inverse image in $O^\#$ is an integral self-dual overlattice for the same positive Riemann form. This gives the claimed principal quotient and pullback equality.

The polarized automorphism group of A is μ_{42}: an O-unit preserving the form satisfies $u\bar{u} = 1$ and is a root of unity. Its image in $\mathbb{F}_{729}^\times/\mathbb{F}_{27}^\times$ has order 7, so it acts freely with four seven-element orbits on the lines. These orbits cannot merge under polarized quotient isomorphisms. Indeed, the quotient lattices are $\pi^{-1}I$ and all have O-saturation $\pi^{-1}O$. Every unmarked isomorphism

between them is scalar by $\mathrm{End}^0(A) = M$. Equality of the saturations forces this scalar to be an O-unit, and preserving their common rational Riemann form forces it into μ_{42}. Conversely every such root induces the corresponding isomorphism. Each fourteen-line B/C orbit splits into two seven-line orbits. Proposition I.7.12 identifies these as the two principal classes on each underlying variety. □

The minimum polarization here differs from the degree-35721 restriction of Proposition 2.2. The quotient statement in Proposition 3.2 concerns the specified $\mathbb{F}_{27}$-lines; the following theorem treats all maximal isotropic subgroups. Neither statement specifies a field of definition or an explicit curve correspondence.

4 Principal and underlying quotient classifications

Theorem 4.1 (All principal targets at minimum degree)**.** *The minimum degree of an isogeny from the actual primitive factor A to any principally polarized abelian variety is* 27*. Exactly* 160 *origin-preserving polarized isomorphism classes of principal targets admit an isogeny of this degree from A.*

Fix $\lambda_{\min}$ from Proposition 3.2. The maps with pullback $\lambda_{\min}$ have exactly 1120 *distinct kernels, the maximal isotropic subspaces of $A[\pi] \simeq \mathbb{F}_3^6$. They give the* 160 *polarized target classes with seven kernels each. Four are B_0, B_1, C_0, C_1, with endomorphism ring R. The remaining* 156 *have full endomorphism ring*

$$S = \mathbb{Z} + \mathfrak{P} \subset \mathcal{O}_M, \qquad [\mathcal{O}_M : S] = 243.$$

Each target has polarized automorphism group μ_6.

Proof. A degree-d map to a principal target pulls back a polarization of degree d^2, so Proposition 3.2 gives $d \geq 27$, attained by the B/C maps. For any degree-27 map, uniqueness of the minimum form gives a source automorphism transporting its pullback to $\lambda_{\min}$. This preserves the polarized target, so it suffices to fix that pullback when counting target classes.

Use the actual lattice $O = \mathcal{O}_M$ and its minimum Riemann form E. Its dual is $O^\# = \pi^{-1}O$. A subgroup $H \subset D = O^\#/O$ gives an integral quotient lattice L_H precisely when it is isotropic, and $L_H^\#/O = H^\perp$. Thus the same positive form is principal on the quotient precisely when $H = H^\perp$. Conversely every map with fixed pullback gives such a self-dual overlattice, so these are exactly its kernels.

If $L_n(q)$ counts Lagrangians in a symplectic space of dimension $2n$ over $\mathbb{F}_q$, counting pairs (v, H) with $0 \neq v \in H$ gives

$$L_n(q)(q^n - 1) = (q^{2n} - 1)L_{n-1}(q), \qquad L_0(q) = 1.$$

Indeed those containing v correspond to Lagrangians in $v^\perp/\langle v \rangle$. Consequently $L_3(3) = (3+1)(9+1)(27+1) = 1120$.

Under multiplication by π, view H as a subspace of $K_{\mathrm{res}} = O/\mathfrak{P} = \mathbb{F}_{729}$, and write $L_H = \pi^{-1}I_H$, where I_H is its inverse image in O. Every nonzero H gives $I_H O = O$, since its nonzero residues generate the field K_{res} as an ideal. All quotient lattices therefore have the same O-saturation $\pi^{-1}O$. Every unmarked quotient isomorphism is scalar in M, by conjugating with the rational quotient maps and using $\mathrm{End}^0(A) = M$. Equality of saturations forces the scalar into $O^\times$; preserving the common form E forces $u\bar{u} = 1$, hence $u \in \mu_{42}$. Conversely such roots give the corresponding quotient isomorphisms. The target classes are exactly the μ_{42}-orbits of Lagrangians.

The residue image of μ_{42} has order 14, with scalar subgroup $\{\pm 1\}$, so the effective action has order 7. A nonidentity effective element is represented, up to sign, by a primitive seventh root. It

generates $\mathbb{F}_{729}$ over $\mathbb{F}_3$, since $\mathrm{ord}_7(3) = 6$. It cannot preserve a three-dimensional subspace: such a subspace would then be a nonzero proper vector subspace over the whole field. The action is free, giving $1120/7 = 160$ classes, and each full stabilizer is the six-element scalar kernel μ_6.

Finally, a multiplier $cI_H \subseteq I_H$ lies in O after extending to O. Its residue must belong to

$$F_H = \{a \in K_{\mathrm{res}} : aH \subseteq H\}.$$

This is a subfield: a nonzero member acts bijectively on finite H, so its inverse also preserves H. Its degree over $\mathbb{F}_3$ divides $\dim_{\mathbb{F}_3} H = 3$. Hence F_H is either $\mathbb{F}_3$ or $\mathbb{F}_{27}$, the latter exactly for the 28 one-dimensional $\mathbb{F}_{27}$-subspaces. Their four polarized classes are the B/C objects already classified. All other 1092 Lagrangians have multiplier equal to the inverse image of $\mathbb{F}_3$, namely $S = \mathbb{Z} + \mathfrak{P}$, of index $729/3 = 243$. They give $1092/7 = 156$ classes. Full rational endomorphisms remain M by isogeny, so these multiplier orders are the full integral endomorphism rings. □

The 1120 kernel count fixes the pullback polarization; the 160 target count allows arbitrary minimum pullbacks. The smaller order S contains $\mathbb{Z}[z^{14}] = \mathcal{O}_{\mathbb{Q}(\sqrt{-3})}$, since $z^{14} \equiv 1 \pmod{\mathfrak{P}}$, but does not contain R. Thus these 156 objects lie outside the selected-order hypothesis of Theorem 12.1. The census does not identify new points on the normalized McMullen curve or classify the new targets as Jacobians.

Theorem 4.2 (Galois orbits of the minimum principal targets)**.** *The* 160 *polarized classes of Theorem 4.1 form eleven orbits under* $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$*. One consists of* B_0, B_1, C_0, C_1*. The other ten, all of full order* S*, have sizes*

$$4, 8, 12, 12, 12, 12, 24, 24, 24, 24.$$

In particular, the property of being the canonically polarized Jacobian of a smooth curve for the 156 *smaller-order classes can be decided on ten representatives.*

Proof. First the literal primitive factor A has a model over $\mathbb{Q}$. The curve $y^2 = x^{21} - 1$ is defined over $\mathbb{Q}$, and Galois sends its Jacobian rotation T to T^a. The primitive cyclotomic projector is invariant under this substitution, which permutes primitive characters. After clearing denominators it is a Galois-invariant homomorphism, hence descends to $\mathbb{Q}$, as does its connected image. All endomorphisms of A are defined over M, since T generates $\mathcal{O}_M$; their Galois action is the cyclotomic action.

Fix a minimum polarization λ. Every conjugate is another minimum polarization on A, so Proposition 3.2 gives an isomorphism $u_\sigma : (A, \sigma\lambda) \longrightarrow (A, \lambda)$, unique modulo μ_{42}. This does not assert that λ descends to $\mathbb{Q}$. The maps $u_\sigma\sigma$ act on $D = \ker\lambda \simeq K_{\mathrm{res}} = \mathbb{F}_{729}$ modulo the residue group μ_{14}. They induce the actual Galois action on the polarized quotient classes: quotients and their polarizations commute with base change, and Theorem 4.1 identifies all polarized quotient isomorphisms with source root actions.

The discriminant pairing has the form

$$b(x, y) = \mathrm{Tr}_{K_{\mathrm{res}}/\mathbb{F}_3}(cxy^{27}), \qquad c^{27} = -c \neq 0.$$

This follows from trace nondegeneracy, adjoint conjugation and alternation. Its semilinear similitude group is

$$G = \{x \mapsto ux^{3^j} : u^{28} = \varepsilon c^{3^j-1}, \quad j \in \mathbb{Z}/6,\ \varepsilon \in \mathbb{F}_3^\times\}.$$

The norm $K_{\mathrm{res}}^\times \to \mathbb{F}_{27}^\times$ is surjective with kernel C_{28}, giving an exact sequence

$$1 \longrightarrow C_{28} \longrightarrow G \longrightarrow C_6 \times C_2 \longrightarrow 1.$$

Thus $\overline{G} = G/\mu_{14}$ has order 24. For $\sigma(\zeta_{42}) = \zeta_{42}^a$, the normalized action has field exponent $3^j \equiv a \pmod 7$ and pairing multiplier $\varepsilon \equiv a \pmod 3$, by Galois functoriality of the polarization pairing. Consequently its image in $\overline{G}$ surjects onto $C_6 \times C_2$.

To obtain the remaining kernel of order two, use CM reciprocity on the maximal-order source. The actual type $\Phi = \{1, 5, 11, 13, 17, 19\}$ has trivial stabilizer; its reflex field is M and its reflex type is the inverse type [11, Example 1.19, Proposition 1.26]. For an idele s supported in the local units at $\mathfrak{P}$, with residue $r \in K_{\mathrm{res}}^\times$, the reflex norm has residue

$$r^{1+3+9+27+243+3} = r^{286}.$$

Since $\gcd(286, 728) = 26$, these residues exhaust C_{28}. Choose σ fixing M with $\sigma|_{M^{\mathrm{ab}}} = \mathrm{art}_M(s)$, using Milne's reciprocal Artin convention. The fundamental theorem supplies an M-quasi-isogeny $\alpha : A \to \sigma A$ with $\alpha N_\Phi(s) = \sigma$ on the finite rational Tate module [11, Theorem 9.10]. The reflex norm is an adelic unit, so α and its inverse are integral on every Tate module and therefore are actual isomorphisms. The polarization multiplier is $\chi_{\mathrm{cyc}}(\sigma)/N_{M/\mathbb{Q}}(s)$ [11, Remark 9.11(c)]. By Lemma 9.5 of that source it is a positive rational number; both numerator and denominator are adelic units, so it equals one. Hence α is polarized. Taking $u_\sigma = \alpha^{-1}$ realizes C_{28}/μ_{14} in the normalized Galois image. That image is therefore all of $\overline{G}$.

The exact finite action on the adopted 1120 isotropic subspaces, grouped into 160 root orbits, gives the eleven asserted orbits. Certificate `s_order_galois_v163` retains the actual discriminant form and coordinate map; a separate implementation enumerates semilinear maps in those coordinates and verifies the same partition. No numerical periods enter this step. Finally, Galois conjugation preserves canonical Jacobians of smooth curves, in both directions, proving the last assertion. □

This orbit theorem does not decide any of the ten remaining Jacobian questions. It permits a separately certified exclusion on one representative to apply to its orbit; it does not identify theta values or torsion labels between distinct normalized period matrices.

Theorem 4.3 (Underlying quotients and the principal obstruction)**.** *The* 33880 *subgroups* $H \subset A[\pi]$ *of order* 27 *give exactly* 188 *origin-preserving unpolarized isomorphism classes of quotients: two with full endomorphism ring* R*, namely* B, C*, and* 186 *with full endomorphism ring* $S = \mathbb{Z} + \mathfrak{P}$*. Exactly* 80 *underlying varieties admit principal polarizations:* B, C *and* 78 *of the* S *varieties. Each admits exactly two principal polarization classes under its full automorphism group. The other* 108 *varieties admit no principal polarization.*

*Allowing arbitrary minimum pullback forms, the degree-*27 *isogenies from fixed* A *to principally polarizable targets have exactly* 14224 *distinct kernels:* 14 *for each* R *target and* 182 *for each of the* 78 *eligible* S *targets. Fixing* $\lambda_{\min}$ *instead gives the* 1120 *kernels of Theorem 4.1, with* 14 *per underlying target, seven for each principal polarization class.*

Proof. Put $k = \mathbb{F}_{27} \subset K_{\mathrm{res}} = \mathbb{F}_{729}$. The real unit t has residue with irreducible polynomial $X^3 - X^2 + X + 1$. Its norm to $\mathbb{F}_3$ is -1, so its order is 26: it does not divide 13, and irreducibility excludes orders $1, 2$. Thus E_L maps onto $k^\times$. The full global-unit image modulo $k^\times$ has order 14, as in the conductor calculation above. Consequently the full image U of $\mathcal{O}_M^\times$ in $K_{\mathrm{res}}^\times$ has order 364 and contains $k^\times$.

Counting ordered bases gives

$$\#\,\mathrm{Gr}(3, 6)(\mathbb{F}_3) = \frac{(3^6 - 1)(3^6 - 3)(3^6 - 9)}{(3^3 - 1)(3^3 - 3)(3^3 - 9)} = 33880.$$

The common saturation and scalar-isomorphism argument in Theorem 4.1 applies to all these subspaces, without imposing isotropy. Unpolarized classes are exactly U-orbits. The stabilizer is $k^\times$ for the 28 k-lines and $\mathbb{F}_3^\times$ otherwise. Hence the orbit sizes are 14 and 182, giving $28/14 = 2$ classes with ring R and $(33880 - 28)/182 = 186$ with ring S.

Set $E_0 = \{u \in E_L : u \bmod \mathfrak{P} \in \mathbb{F}_3^\times\}$. Then $[E_L : E_0] = 13$. This odd index preserves the signature image: $u^{13} \in E_0$ has the same signs as u. The image therefore has order 32, whereas E_0/E_0^2 has order 64 by its rank five and torsion $\{\pm 1\}$. Thus $[E_0^+ : E_0^2] = 2$. Furthermore $S^\times = \mu_6 E_0$. Indeed, for $u \in S^\times$, $u/\bar{u}$ is a root of unity reducing to 1, hence belongs to μ_3; dividing by a suitable element of μ_6 makes u real without changing the scalar residue condition. It follows that $N_{M/L}(S^\times) = E_0^2$.

For any polarization on an S quotient, pullback through the scalar isogeny from A gives a polarization on A. Its Rosati involution on M is conjugation by Proposition 3.2; transport through the rational isogeny proves the same for the quotient. If an S variety has one principal form λ_0, every other principal form is $\lambda_0 a$ with $a \in E_0^+$: symmetry is complex conjugation on M, and positivity is total positivity at its real places. Its full unmarked automorphism group is $S^\times$, so the classes are E_0^+/E_0^2, exactly two. The two R varieties also have two principal classes each by Proposition I.7.12.

Any principal form on any quotient pulls back to a degree-729 polarization on A. Proposition 3.2 transports it to $\lambda_{\min}$ by a source automorphism. Thus Theorem 4.1 exhausts all principal polarization classes on all these quotients, giving $160/2 = 80$ underlying varieties and $188 - 80 = 108$ absolute principal obstructions.

For any minimum map to a principally polarizable target, the same source normalization puts its kernel in $A[\pi]$. Source units commute with π, so its original kernel is also in $A[\pi]$. Conversely, every subgroup whose quotient is principally polarizable gives such a minimum map. Orbit sizes now give $2 \cdot 14 + 78 \cdot 182 = 14224$ kernels; the other $108 \cdot 182 = 19656$ subgroups have nonprincipalizable quotients. For the fixed pullback, each of the two principal classes contributes seven kernels. They cannot share a kernel, since the common rational form determines the quotient polarization uniquely. □

These counts concern complex abelian varieties and origin-preserving maps. They neither identify new points of the normalized McMullen curve nor assert a Jacobian classification for the S varieties.

Corollary 4.4 (Galois orbits after forgetting polarization)**.** *The* 80 *underlying principalizable varieties of Theorem 4.3 form eleven Galois orbits. One consists of B, C; the ten smaller-order orbits have sizes*

$$2, 4, 6, 6, 6, 6, 12, 12, 12, 12.$$

Each orbit in Theorem 4.2 maps onto exactly one of these underlying orbits, with fibers of size two.

Proof. Write U_d for the subgroup of order d in $K_{\mathrm{res}}^\times$. The subgroup U_{28}/U_{14}, the kernel of the projection $\overline{G} \to C_6 \times C_2$ in the normalized Galois group, acts trivially on underlying varieties, because $U_{28} \subset U_{364}$ and U_{364} is the full global-unit residue image in the proof of Theorem 4.3. The unit lift gives an actual lattice isomorphism.

This kernel acts freely on polarized classes. A fixed class would give $uH = vH$ with $u \in U_{28} \setminus U_{14}$ and $v \in U_{14}$. The scalar $v^{-1}u$ has order 4 or 28, so generates a field of degree 2 or 6 over $\mathbb{F}_3$. The invariant subspace H would be a vector space over that field, contradicting $\dim_{\mathbb{F}_3} H = 3$. Each forgetful fiber already has exactly two polarized classes, hence is exactly one free kernel orbit. Equivariance prevents distinct polarized Galois orbits from merging and halves every orbit size, as claimed. The human-proof certificate is `principalizable_galois_corollary_v164`. □

This corollary concerns the principalizable subfamily only. It does not classify Galois orbits of the 108 nonprincipalizable varieties or assert descent to a field of moduli.

Proposition 4.5 (Cyclotomic action on all underlying quotients)**.** *On all* 188 *underlying classes of Theorem 4.3, the Galois action factors faithfully through* $\mathrm{Gal}(M/\mathbb{Q}) \simeq C_6 \times C_2$*. Its orbits are exactly the full semilinear orbits on the three-dimensional* $\mathbb{F}_3$*-subspaces of* $K_{\mathrm{res}} = \mathbb{F}_{729}$*, modulo the global-unit subgroup* U_{364}*. Every orbit size divides* 12*. The principalizable and nonprincipalizable subsets are invariant.*

Proof. The rational source A and intrinsic subgroup $A[\mathfrak{P}]$ are Galois-invariant, as in Theorem 4.2. For every subgroup H, base change identifies the conjugate quotient with $A/\sigma H$, and the actual normalizing unit u_σ identifies this with $A/(u_\sigma \sigma H)$. This requires no isotropy. The common-saturation argument of Theorem 4.3 identifies all underlying classes with U_{364}-orbits. Choice ambiguities lie in $U_{14} \subset U_{364}$. The proof of Theorem 4.2 establishes surjectivity onto the group of semilinear maps G/U_{14} before its isotropic orbit calculation, so applies on this full domain.

The subgroup U_{364} is normal in $\Gamma = K_{\mathrm{res}}^{\times} \rtimes \mathrm{Gal}(K_{\mathrm{res}}/\mathbb{F}_3)$. For a generator g of $K_{\mathrm{res}}^{\times}$, the scalar g^{13} lies in G, since $(g^{13})^{28} = -1$. Together with $U_{364} = \langle g^2 \rangle$ it generates all scalars. Since G surjects onto the six field automorphisms, $\langle U_{364}, G \rangle = \Gamma$. The quotient $\Gamma/U_{364} \simeq C_2 \times C_6$ has order 12.

If σ fixes M, its normalized action has $j = 0$ and $\epsilon = 1$, hence scalar in $U_{28} \subset U_{364}$. The actual action therefore factors through $\mathrm{Gal}(M/\mathbb{Q})$. Corollary 4.4 supplies an orbit of size 12; its point stabilizer in this order-12 group is trivial, proving faithfulness. Principalizability is preserved by conjugation and its inverse. The human-proof certificate is `all_underlying_galois_v165`. □

This determines the acting group on the 108 nonprincipalizable classes, but does not enumerate their orbits or prove descent of individual varieties. There are 94 full-scalar orbits on subspaces: one of size 28 and 93 of size 364, by the multiplier-field stabilizers above. Frobenius may merge these scalar orbits.

5 Minimum forms on nonprincipalizable quotients

Theorem 5.1 (Minimum polarizations on the obstructed quotients)**.** *Each of the* 108 *nonprincipalizable varieties of Theorem 4.3 has minimum polarization degree* $3^{12} = 531441$*, with type* $(1, 3, 3, 3, 3, 9)$*. Each has exactly* 26 *minimum polarization classes under its full origin-preserving automorphism group.*

Proof. Write $O = \mathcal{O}_M$, let $\mathfrak{Q} = \mathfrak{P} \cap \mathcal{O}_L$, and fix the positive minimum form E on O, with trace parameter $\kappa_O a_0$. Proposition 3.2 gives $(a_0) = \mathfrak{Q}$, $N_{L/\mathbb{Q}}(a_0) = -27$, $\mathfrak{Q}O = \mathfrak{P}$, and $(3) = \mathfrak{Q}^2$ in $\mathcal{O}_L$. For $Y = A/H$, let Λ be the inverse image of H in $\mathfrak{P}^{-1}$. Then $[\Lambda : O] = 27$ and $\Lambda O = \mathfrak{P}^{-1}$.

Pulling any polarization back to A and transporting its Rosati involution shows that every polarization on Y has trace parameter $\kappa_O a$ with $a \in L$. Integrality on $O \times O$ gives $a \in \mathcal{O}_L$. Mixed integrality on $O \times \Lambda$ is equivalent to $a\Lambda \subset O$ by trace duality and conjugation. Taking O-saturations gives $a \in \mathfrak{P} \cap L = \mathfrak{Q}$; conversely this condition suffices for mixed integrality. Thus $a = a_0 v$ with $v \in \mathcal{O}_L$, and positivity is precisely total positivity of v.

On $D = \mathfrak{P}^{-1}/O$ define, for $c \in k = \mathbb{F}_{27}$,

$$b_c(x + O, y + O) = 3E(vx, y) \pmod 3, \qquad v \bmod \mathfrak{Q} = c.$$

This is independent of the lift and representatives: $E(O,\mathfrak{P}^{-1}) \subset \mathbb{Z}$ and $\mathfrak{Q}\mathfrak{P}^{-1} \subset O$. The forms are alternating, and b_c is nondegenerate for $c \neq 0$. Put

$$T_H = \{c \in k : b_c|_{H\times H} = 0\}.$$

This is an $\mathbb{F}_3$-linear subspace. Integrality on $\Lambda \times \Lambda$ is equivalent to $v \bmod \mathfrak{Q} \in T_H$. For an integral positive form $E_v(x,y) = E(vx,y)$, its degree on Y is

$$\deg(E_v|_\Lambda) = \frac{729 N_{L/\mathbb{Q}}(v)^2}{[\Lambda : O]^2} = N_{L/\mathbb{Q}}(v)^2.$$

This determinant calculation does not assume that E itself is integral on Λ.

By Theorem S.11.3, the residue image of E_L^+ is the norm of the order-364 group U from Theorem 4.3. Conjugation on K_{res} is the 27th power map, so this norm image has order $364/\gcd(364,28) = 13$, exactly the squares in $k^\times$. Since -1 is a nonsquare, every nonzero pair $\{c,-c\}$ contains such a residue. If $T_H \neq 0$, linearity therefore gives a totally positive unit ε with residue in T_H. Then E_ε is a principal form on Y. Conversely a principal form has a unit parameter and hence $T_H \neq 0$. For the 108 obstructed varieties, $T_H = 0$, and all positive integral parameters are exactly the totally positive elements of $\mathfrak{Q}$.

Write $\mathfrak{Q} = (\beta)$ with $N_{L/\mathbb{Q}}(\beta) = -27$. For such a parameter $v = \beta w$, positivity gives $N(w) < 0$. The signed norm bound of Proposition 3.2 gives $|N(w)| \geq 27$, so $N(v) \geq 729$. Equality is attained by $v = 3$. At equality (v) has norm 3^6, hence is $\mathfrak{Q}^2 = (3)$, since $\mathfrak{Q}$ is the unique prime above 3, of norm 27. Thus all minimum parameters are exactly $3E_L^+$, and the minimum degree is 729^2.

The full automorphism group is $S^\times$, with norm image E_0^2 in the notation of Theorem 4.3. The equal signature images of E_0 and E_L give $[E_L^+ : E_0^+] = 13$. The minimum classes are therefore E_L^+/E_0^2, of size $13 \cdot 2 = 26$.

Finally let $F = 3E_\varepsilon$ be a minimum form. Modulo 3 it vanishes on $O \times \Lambda$ and factors through $H = \Lambda/O$, since $3\Lambda \subset O$. The induced alternating form is $b_{\varepsilon \bmod \mathfrak{Q}}|_H$, nonzero because $T_H = 0$. It has rank 2 on the three-dimensional space H. Thus exactly one of the six elementary divisors of F is prime to 3. Their product is 3^6, so that divisor is 1 and the other five have positive 3-adic exponents summing to 6. The type is $(1,3,3,3,3,9)$. □

The residue obstruction also gives a direct count. If H is isotropic for two independent coefficients $c,d \in k$, then it is a Lagrangian for b_c and stable under d/c. This ratio generates k over $\mathbb{F}_3$, so H is a k-line. Each of the 28 such lines is isotropic for every b_c. The 13 coefficient lines therefore have disjoint non-k-line Lagrangian populations, giving $28 + 13(1120 - 28) = 14224$ principalizable kernels and 19656 obstructed ones. This finite incidence identity corroborates the full-unit orbit classification; the polarization minima additionally require the signed norm argument above.

Proposition 5.2 (Normalized group and minimum polarized action)**.** *The normalized group $\overline{G}$ of Theorem 4.2 is $D_8 \times C_3$, with D_8 of order 8. Its commutator is the central involution U_{28}/U_{14}, and its abelianization is $C_6 \times C_2$.*

The $2808 = 108 \cdot 26$ minimum polarized classes on the obstructed quotients identify with the U_{14}-orbits of the 19656 subspaces with $T_H = 0$, using the form descended from $3\lambda_{\min}$. Their actual Galois action is the restriction of this same normalized group, so every orbit size divides 24. The central involution acts freely and preserves the underlying variety. Each 26-element forgetful fiber is therefore the union of 13 free pairs.

Proof. The scalar subgroup of G is U_{56}; its quotient by U_{14} is cyclic of order 4, generated by a. Choose $b \in \overline{G}$ above the Frobenius generator of C_6. Then $bab^{-1} = a^3 = a^{-1}$. The cube of a

representative is $w\,\mathrm{Frob}^3$ with multiplier $\epsilon \in \{1,-1\}$. Its square is the scalar

$$w^{28} = \epsilon c^{26} = -\epsilon \in U_{14},$$

since $c^{27} = -c$. Thus $b^6 = 1$, and its projection forces its order to be 6. The subgroups generated by a and b intersect trivially and generate $\overline{G}$. Now b^2 is central of order 3, whereas a, b^3 generate D_8. These commuting factors intersect trivially. Their commutator is $\langle a^2\rangle = U_{28}/U_{14}$, proving the group assertions.

By Theorem 5.1, all minimum forms on an obstructed lattice are $3E_\varepsilon$ with $\varepsilon \in E_L^+$. Choose $u \in \mathcal{O}_M^\times$ with $u\bar{u} = \varepsilon^{-1}$ using Theorem S.11.3. Multiplication by u from $\Lambda_{u^{-1}H}$ to Λ_H pulls $3E_\varepsilon$ back to $3E$. Conversely $3E$ is integral on every such lattice and is minimum when $T_H = 0$. A fixed-pullback polarized isomorphism between two quotients is a global unit v by their common $\mathcal{O}_M$-saturation. Preserving $3E$ gives $v\bar{v} = 1$; Kronecker's theorem then gives $v \in \mu_{42}$, with residue in U_{14}. This proves the claimed equivalence relation.

An order-7 scalar generates $\mathbb{F}_{729}$ over $\mathbb{F}_3$ and cannot preserve a three-dimensional subspace. Every root orbit therefore has size 7, giving $19656/7 = 2808$. The source units normalizing $\lambda_{\min}$ also normalize $3\lambda_{\min}$; quotients and descended forms commute with base change. Hence Theorem 4.2 supplies the actual action here. Nonprincipalizability is invariant under conjugation.

If $u \in U_{28} \setminus U_{14}$ fixes a root orbit, then $uH = vH$ for some $v \in U_{14}$. The scalar $v^{-1}u$ has order 4 or 28, and generates a field of degree 2 or 6 over $\mathbb{F}_3$. Its invariant subspace cannot have dimension 3. Thus the involution acts freely. Since $U_{28} \subset U_{364}$, it preserves the underlying quotient, proving the fiber assertion. The human-proof certificate is `obstructed_minimum_galois_v165`. □

The 13 pairs do not exhaust the forgetful fiber as a single kernel orbit. Distinct polarized Galois orbits can merge after forgetting the polarization; the division-by-two orbit conclusion of Corollary 4.4 does not extend to these 26-element fibers. No orbit list or descent statement follows from this proposition alone.

6 Galois action fields and orbit classification

Theorem 6.1 (The ray field of the polarized-class action)**.** *Let $\mathfrak{P}$ be the unique prime of M above 3, and put*

$$\mathcal{E} = M\left(\sqrt{\frac{3+\sqrt{21}}{2}}\right).$$

Then $\mathcal{E}$ is the ray class field of M of modulus $\mathfrak{P}$, with $[\mathcal{E} : M] = 2$ and relative discriminant and conductor $\mathfrak{P}$. It is Galois over $\mathbb{Q}$, with group $D_8 \times C_3$, where $|D_8| = 8$.

The kernel of the normalized action on the 160 principal polarized classes is exactly $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathcal{E})$. Their fields of moduli are contained in $\mathcal{E}$, with degrees equal to their orbit sizes; a class in a size-24 orbit has field of moduli $\mathcal{E}$. The 2808 obstructed minimum polarized classes also have fields of moduli contained in $\mathcal{E}$. These statements do not assert effective descent of polarized models. For the complete finite family, including A, the common fixing fields are M for the 189 underlying classes and $\mathcal{E}$ for the 2969 minimum polarized classes.

Proof. The quadratic subfields $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-7})$ of M give $F = \mathbb{Q}(\sqrt{21}) \subset M$. Put $\omega = (1+\sqrt{21})/2$ and $a = \omega + 1$. Then $\omega^2 - \omega - 5 = 0$, $\omega^2 - 4 = a$ and $N_{F/\mathbb{Q}}(a) = -3$. In F, (a) is the prime above 3. Since 3 has order 6 modulo 7, the prime in M has ramification index 2 and residue degree 6. Its ramification index over F is 1, so $v_{\mathfrak{P}}(a) = 1$ and $(a)\mathcal{O}_M = \mathfrak{P}$.

Let t satisfy $t^2 - \omega t + 1 = 0$. The integral order $\mathcal{O}_M[t]$ has discriminant a, which is a unit away from $\mathfrak{P}$, including above 2. At $\mathfrak{P}$, its discriminant valuation 1 precludes a nontrivial local index. Thus $\mathcal{E} = M(t)$ is quadratic over M with discriminant $\mathfrak{P}$. The quadratic conductor–discriminant formula gives the same conductor, with no infinite component because M is CM [10, Theorem V.3.27]. For an explicit equation, $\omega = t + t^{-1}$ gives the minimal polynomial

$$t^4 - t^3 - 3t^2 - t + 1 = 0$$

over $\mathbb{Q}$: $F \subset \mathbb{Q}(t)$ and a is nonsquare even in M, so the degree is 4.

The class number of M is 1; independently, this follows from Masley's Main Theorem for $m = 21$, with $\varphi(21) = 12 \leq 24$ and 21 absent from its exceptional table [5, p. 179]. The full-unit residue image is U_{364} by Theorem 4.3. The finite-modulus ray sequence therefore gives

$$\mathrm{Cl}_{\mathfrak{P}}(M) = \mathbb{F}_{729}^{\times}/U_{364} \simeq C_2.$$

Ray class reciprocity and existence identify the conductor-$\mathfrak{P}$ quadratic extension just constructed with the ray field [10, Theorems V.1.7, V.3.6 and Remark V.3.8].

Restrict the normalized homomorphism of Theorem 4.2 to $\mathrm{Gal}(\overline{\mathbb{Q}}/M)$. Its image is $U_{28}/U_{14} \simeq C_2$, hence it is a continuous abelian character. For a local unit at a finite place v, form the idele equal to that unit at v and 1 elsewhere. Its reflex norm is an everywhere integral adelic unit. The same Tate-lattice and cyclotomic-factor argument used in Theorem 4.2 makes the CM comparison map an actual polarized isomorphism. Consequently the normalized residue action is the reflex norm modulo U_{14}.

If $v \neq \mathfrak{P}$, the reflex norm has component 1 at $\mathfrak{P}$: automorphisms preserve the underlying rational prime, and $\mathfrak{P}$ is the only place above 3. At $\mathfrak{P}$ the residue is r^{286}, as computed in Theorem 4.2. This is trivial on principal units and maps the full residue group onto U_{28} because $\gcd(286, 728) = 26$. The quadratic character has conductor exactly $\mathfrak{P}$, by the local unit-filtration criterion [10, Section I.1.9]; its fixed extension of M is therefore $\mathcal{E}$.

The quotient of the normalized action by U_{28}/U_{14} is precisely $\mathrm{Gal}(M/\mathbb{Q})$, so its full kernel is already contained in $\mathrm{Gal}(\overline{\mathbb{Q}}/M)$. It equals $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathcal{E})$. Surjectivity and Proposition 5.2 give the claimed degree and group over $\mathbb{Q}$. Faithfulness on the 160 classes identifies their common fixing field exactly. The 2808-class action factors through the same normalized map. Taking stabilizer fixed fields proves the fields-of-moduli assertions. No descent datum is supplied by this stabilizer calculation. Finally A is defined over $\mathbb{Q}$ and has a unique minimum polarized isomorphism class, by Proposition 3.2. Adding these fixed classes changes neither kernel. Proposition 4.5 gives the underlying common field M; the faithful 160-class subset detects the full polarized kernel while the 2808-class action factors through it. This proves the complete-family assertion. □

Theorem 6.2 (Complete Galois orbit census of the quotient family)**.** *The* 188 *underlying quotient classes have* 22 *Galois orbits, with the following sizes and multiplicities:*

orbit size	2	4	6	12
number of orbits	2	4	4	12.

The 108 *obstructed underlying classes form three orbits of size* 4 *and eight of size* 12*. Their* 2808 *minimum polarized classes form three orbits of size* 8 *and* 116 *of size* 24*. Above each underlying size-*4 *orbit lie one polarized size-*8 *orbit and four size-*24 *orbits; above each size-*12 *orbit lie thirteen size-*24 *orbits.*

Including A*, the complete family has* 23 *underlying orbits and* 131 *minimum polarized orbits. The latter consist of one orbit of size* 1*, two of size* 4*, four of size* 8*, four of size* 12*, and* 120 *of size* 24*.*

Proof. Write $K = \mathbb{F}_{729}$, $k = \mathbb{F}_{27}$ and $F(x) = x^3$. Let X be the 33880 three-dimensional $\mathbb{F}_3$-subspaces of K. We first count invariant subspaces of $T = uF^j$. For $d = \gcd(6, j)$, with $d = 6$ if $j = 0$, put $n = N_{K/\mathbb{F}_{3^d}}(u)$. Scalar conjugation changes u by v^{1-3^j}, whose image is precisely the norm-one subgroup. Thus the fixed-space count depends only on n, and each norm value occurs $728/(3^d - 1)$ times.

For $j = 0$, invariance makes a three-plane a vector space over $\mathbb{F}_3(u)$, so only degrees 1 and 3 contribute. The scalars ± 1 fix all 33880 planes, the other 24 elements of $k^\times$ fix the 28 k-lines, and all others fix none.

For $j = 1, 5$, the operator is cyclic with minimal polynomial $X^6 - n$. Indeed, a shorter relation among its powers would contradict linear independence of the six field automorphisms over K. For $n = 1$, the two primary cyclic blocks $(X - 1)^3, (X + 1)^3$ have four submodules of total dimension 3. For $n = -1$, the single irreducible quadratic factor of $(X^2 + 1)^3$ permits only even dimensions.

For $j = 2, 4$, a norm value outside $\mathbb{F}_3$ forces invariance under $\mathbb{F}_9$, again excluding dimension 3. For $n = \pm 1$, a sign and scalar conjugation reduce to F^j. The nilpotent operator $F^j - 1$ has cube zero and two-dimensional kernel, hence Jordan type $(3, 3)$. Its invariant three-planes number

$$\frac{3^6 - 3^4}{3^3 - 3^2} + \binom{2}{1}_3 = 36 + 4 = 40.$$

Here the cyclic submodules have length 3; those of type $(2, 1)$ contain the kernel and correspond to lines of the second kernel modulo the kernel. Type $(1, 1, 1)$ is impossible.

For $j = 3$, T is k-linear with $T^2 = n$. If $n = 1$, its two eigenspaces have dimension 3 over $\mathbb{F}_3$, giving

$$\sum_{r=0}^{3} \binom{3}{r}_3 \binom{3}{3-r}_3 = 1 + 169 + 169 + 1 = 340$$

invariant three-planes. If $n = -1$, an $\mathbb{F}_9$-structure excludes them. Each of the twelve nontrivial square values in $k^\times$ has degree 3 and gives two invariant three-planes, its eigenspaces. The eigenvalues $r, -r$ are not Frobenius conjugate: $r^{3^i - 1} = -1$ is impossible in the order-26 group for $i = 1, 2$, since the power image has odd order. The remaining twelve nonsquare values generate a degree-6 field after taking a square root and give no invariant three-plane.

Let $S = X/K^\times$, of size 94. The coset form of Burnside's lemma says that the number of F^j-fixed elements of S is the average of $|\operatorname{Fix}_X(uF^j)|$ over $u \in K^\times$. The preceding counts give

j	0	1	2	3	4	5
$\lvert\operatorname{Fix}_S(F^j)\rvert$	94	2	10	14	10	2.

For example the three nontrivial averages are $364 \cdot 4/728 = 2$, $91 \cdot 2 \cdot 40/728 = 10$, and $28(340 + 12 \cdot 2)/728 = 14$. Fixed-point inversion gives respectively $2, 4, 4, 12$ Frobenius orbits of lengths $1, 2, 3, 6$. Every full-scalar orbit splits into two U_{364}-orbits: its stabilizer is either $\mathbb{F}_3^\times$ or $k^\times$, both contained in U_{364}. Thus these lengths double in the 188-class action of Proposition 4.5. Subtracting Corollary 4.4 gives the obstructed underlying distribution.

For the polarized count, use the actual similitude group $G = \{uF^j : u^{28} = \epsilon c^{3^j - 1},\ \epsilon = \pm 1\}$, with $c^{27} = -c$ and $|G| = 336$. The sums of fixed three-planes restricted to these maps are

j	0	1	2	3	4	5
$\sum_{uF^j \in G} \lvert\operatorname{Fix}_X(uF^j)\rvert$	67760	112	560	9520	560	112.

To verify the restriction, at $j = 0$ only ± 1 contribute. For odd $j = 1, 5$, $u^{364} = -\epsilon$, so the 28 maps with $\epsilon = -1$ each fix four planes. For $j = 2, 4$, the four possible values $n = u^{91}$ satisfy $n^4 = \epsilon$, each

with multiplicity 7; only $n = \pm 1$ contribute, giving $2 \cdot 7 \cdot 40$. At $j = 3$, $n = u^{28} = -\epsilon$, and the 28 involutions each fix 340 planes. Burnside therefore gives 234 G-orbits on X.

We remove the principalizable planes by incidence. The thirteen coefficient lines in k have pullback substitution $t \mapsto t^{3^{-j}}$, up to the irrelevant scalar ϵ. Forward incidence transport uses its inverse; the fixed lines, stabilizers and orbit partition are the same. They form one fixed line $\mathbb{F}_3$ and four orbits of length 3. A line in a nontrivial orbit has stabilizer $G_0 = \{uF^j \in G : j = 0, 3\}$, of order 112. Scalar multiplication identifies its Lagrangians with those for the actual form and normalizes G_0: at $j = 3$ it changes u by v^{26}, whose 28th power is 1.

On the 1120 Lagrangians, the $j = 0$ fixed sum for G_0 is $2 \cdot 1120$. At $j = 3$, the maps with $\epsilon = 1$ square to -1 and fix no three-plane. Each of the 28 maps with $\epsilon = -1$ is an anti-symplectic involution with complementary Lagrangian eigenspaces. An invariant Lagrangian is a subspace of one eigenspace together with its annihilator in the other, giving $\sum_r \binom{3}{r}_3 = 28$ choices. Hence G_0 has $(2240 + 28 \cdot 28)/112 = 27$ Lagrangian orbits.

Consider pairs (H, ℓ) with $\ell \subset T_H$ a coefficient line. Their number of G-orbits is $11+4\cdot 27 = 119$, using Theorem 4.2 for the fixed coefficient line. A generic principalizable plane occurs once; a k-line occurs thirteen times. The 28 k-lines are transitive under G and under its normal index-3 subgroup G_0, since the number of G_0-orbits divides both 3 and 28. The incidence over these lines has five orbits, while the lines themselves have one. Thus the principalizable planes have $119 - 5 + 1 = 115$ orbits, leaving $234 - 115 = 119$ obstructed polarized orbits.

The central involution acts freely, so stabilizers in $D_8 \times C_3$ have orders $1, 2, 3, 6$ and orbit sizes $24, 12, 8, 4$. Let h generate the central C_3. Choose a representative g above F^2 with $g^3 = 1$. For each $v \in U_7$, the map $v^{-1}g$ also has cube 1, since $v^{91} = 1$, and fixes 40 three-planes. Their seven fixed sets are disjoint: an overlap would make a three-plane invariant under an order-7 scalar. These 280 planes form exactly 40 root orbits fixed by h.

For a fixed principalizable plane outside the k-lines, its unique coefficient line must be fixed by F^2, so it is the actual coefficient line $\mathbb{F}_3$. The k-lines are also isotropic. Hence the principalizable fixed root orbits are exactly the fixed classes in the 160-class population. Its two size-4 orbits and one size-8 orbit give 16 fixed classes. Thus h fixes $40 - 16 = 24$ obstructed polarized classes. If a_d counts their orbits of size d, then

$$\begin{aligned} a_4 + a_8 + a_{12} + a_{24} &= 119, \\ 4a_4 + 8a_8 + 12a_{12} + 24a_{24} &= 2808, \\ 4a_4 + 8a_8 &= 24. \end{aligned}$$

Nonnegativity gives $a_4 = 6 - 2a_8 \geq 0$ and $a_{12} = 2a_8 - 6 \geq 0$, forcing $a_8 = 3$, $a_{24} = 116$ and $a_4 = a_{12} = 0$.

Each polarized orbit maps onto an underlying orbit. The 312 classes over an underlying size-12 orbit therefore form thirteen size-24 orbits. Over each size-4 orbit the equation $8x + 24y = 104$ forces $x \equiv 1 \pmod 3$. There are three such underlying orbits and only three size-8 polarized orbits in total, so each has $x = 1, y = 4$. Finally add the known principal orbits and the fixed minimum class of A. This proves all counts. □

Corollary 6.3 (Fields of moduli in the obstructed family)**.** *Put $F_0 = \mathbb{Q}(\sqrt{-3}, \sqrt{-7})$ and $E_0 = F_0(\sqrt{(3 + \sqrt{21})/2})$. The obstructed underlying classes in size-4 orbits have field of moduli F_0; those in size-12 orbits have field of moduli M. The minimum polarized classes in size-8 orbits have field of moduli E_0, and those in size-24 orbits have field of moduli $\mathcal{E}$. In particular their action is faithful, with common fixing field $\mathcal{E}$.*

The field E_0 is the degree-8 dihedral splitting field of $X^4 - X^3 - 3X^2 - X + 1$. None of these field-of-moduli statements asserts effective descent.

Proof. The unique order-3 subgroup of $\mathrm{Gal}(M/\mathbb{Q})$ fixes F_0; it is the stabilizer in a size-4 orbit. A size-12 orbit has trivial stabilizer. Similarly the unique central C_3 in $\mathrm{Gal}(\mathcal{E}/\mathbb{Q})$ is the stabilizer in a size-8 orbit. It fixes F_0 and cannot negate the square root of $a = (3+\sqrt{21})/2$, since its order is 3. Thus it fixes E_0. Since $\sqrt{a} \notin M$, the degree of E_0 over $\mathbb{Q}$ is 8, so this is the full fixed field. A size-24 orbit has trivial stabilizer. Apply Theorem 6.1.

Write $a = (3+\sqrt{21})/2$ and $a' = (3-\sqrt{21})/2$, so $aa' = -3$. The field E_0 contains both $\sqrt{a}$ and $\sqrt{a'} = \sqrt{-3}/\sqrt{a}$, hence all four roots t, t^{-1}, t', t'^{-1}. Conversely their sums give $\sqrt{21}$, their differences give $\sqrt{a}, \sqrt{a'}$, and the product of those differences gives $\sqrt{-3}$. The splitting field therefore contains F_0 and $\sqrt{a}$, and equals E_0. □

Theorem 6.4 (Fields of moduli for the complete finite family)**.** *Choose* $\sqrt{21} > 0$*, put* $a = (3+\sqrt{21})/2$*, and set*

$$D = \mathbb{Q}\left(\frac{(1+\sqrt{21})/2 + \sqrt{a}}{2}\right), \qquad J = \mathbb{Q}(\zeta_7 + \zeta_7^{-1}), \qquad M^+ = \mathbb{Q}(\zeta_{21} + \zeta_{21}^{-1}).$$

Retain $F_0 = \mathbb{Q}(\sqrt{-3}, \sqrt{-7})$*,* $E_0 = F_0(\sqrt{a})$ *and* $\mathcal{E} = M(\sqrt{a})$*. The fields of moduli of the complete quotient family, including A, are given by the following tables. Non-normal fields are specified up to* $\mathbb{Q}$*-conjugacy in* $\mathcal{E}$*.*

underlying orbit size	1	2	4	6	12
number of orbits	1	2	4	4	12
field of moduli	$\mathbb{Q}$	$\mathbb{Q}(\sqrt{21})$	F_0	M^+	M

minimum polarized orbit size	1	4	8	12	24
number of orbits	1	2	4	4	120
field of moduli	$\mathbb{Q}$	D	E_0	DJ	$\mathcal{E}$

Here D *has degree* 4 *and signature* $(2,1)$*, while* DJ *has degree* 12 *and signature* $(6,3)$*. With the chosen embedding,* $DJ = \mathcal{E} \cap \mathbb{R}$*; neither* D *nor* DJ *is totally real. These remain fields of moduli, with no assertion of effective descent.*

Proof. The normalized action projects to $\mathrm{Gal}(M/\mathbb{Q})$ by (j, ϵ), where $\zeta_7 \mapsto \zeta_7^{3^j}$ and $\zeta_3 \mapsto \zeta_3^{\epsilon}$. In particular complex conjugation projects to $(3, -1)$. A stabilizer avoiding the free central involution in $D_8 \times C_3$ has order $1, 2, 3$ or 6. Its order-2 factor, if present, is a reflection. The two reflection conjugacy classes are distinguished by the multipliers $\epsilon = 1$ and -1 at $j = 3$: the two lifts over either pair differ by the central involution and are conjugate by an order-4 scalar.

Write $K_{\mathrm{res}} = \mathbb{F}_{729}$ and let $\mathrm{Frob}(x) = x^3$ on K_{res}. A reflection of multiplier 1 fixes no root orbit of any three-plane $H \subset K_{\mathrm{res}}$. Indeed write it as $g = u\,\mathrm{Frob}^3$. Then $g^2 = u^{28} = -\epsilon$. If $gH = vH$ for $v \in U_{14}$, the operator $v^{-1}g$ preserves H and has square $-\epsilon$, because $v^{28} = 1$. For $\epsilon = 1$, its irreducible polynomial $X^2 + 1$ would make H a vector space over $\mathbb{F}_9$, impossible in dimension 3 over $\mathbb{F}_3$. Thus every reflection stabilizer is conjugate to complex conjugation c in $\mathrm{Gal}(\mathcal{E}/\mathbb{Q})$.

It follows that the stabilizers of the principal polarized orbits of sizes $4, 8, 12, 24$ are, up to conjugacy, $C_3 \times \langle c \rangle$, C_3, $\langle c \rangle$, and 1. The obstructed polarized orbits have only sizes $8, 24$, already covered by Corollary 6.3.

The fields E_0 and J lie in $\mathcal{E}$ and have coprime degrees $8, 3$, so $\mathcal{E} = E_0 J$. By the quartic irreducibility in Theorem 6.1, $D = \mathbb{Q}(\sqrt{21}, \sqrt{a})$ has degree 4. It is fixed by c, as is J, and $[DJ : \mathbb{Q}] = 12$. Consequently $\mathcal{E}^{\langle c \rangle} = DJ$ and $\mathcal{E}^{C_3 \times \langle c \rangle} = D$. The quartic has two real roots over $\sqrt{21} > 0$ and one complex pair over its negative conjugate. Since J is totally real and linearly disjoint from D, the signatures follow.

For underlying principalizable classes, forgetting the two forms is exactly quotienting by the free central involution z. Thus a polarized stabilizer H gives the underlying stabilizer $H\langle z\rangle/\langle z\rangle$ in $\mathrm{Gal}(M/\mathbb{Q})$. The size-6 underlying orbits therefore have fixed field M^+; the size-2 orbits have the field fixed by C_3 and complex conjugation, namely $\mathbb{Q}(\sqrt{21})$. The size-4 and size-12 fields are F_0 and M. Add the obstructed underlying orbits from Corollary 6.3 and the rational class of A, with its unique minimum polarization class. The multiplicities are those of Theorem 6.2. □

7 Real models and actual descent

Theorem 7.1 (Real models and pseudoreal underlying varieties)**.** *In the complete* 189*-class underlying family, exactly* 15 *complex isomorphism classes admit a model over* $\mathbb{R}$*. Each admits a model over* $\mathbb{R}_{\mathrm{alg}} = \overline{\mathbb{Q}} \cap \mathbb{R}$*. Another* 14 *classes are* pseudoreal*: they are isomorphic to their complex conjugates but have no model over* $\mathbb{R}$*.*

Exactly 28 *of the* 160 *principal polarized quotient classes admit real models, comprising two* R*-order and* 26 S*-order classes. They lie over* 14 *underlying classes, one of order* R *and* 13 *of order* S*, with both principal classes real on each. The* 14 *pseudoreal classes also comprise one* R*-order and* 13 S*-order classes. Including* A *with its minimum polarization gives* 29 *real minimum polarized classes. None of the* 2808 *obstructed minimum polarized classes has a real model.*

These count complex isomorphism classes admitting models, not inequivalent real forms. Polarizations mean polarization morphisms; no chosen line-bundle linearization or descent to the smallest number field is asserted.

Proof. Write c for the chosen complex conjugation. The literal Fermat factor A is defined over $\mathbb{Q}$, and the restricted Jacobian polarization μ is rational. Every geometric polarization λ on A has $\lambda = \mu\alpha$ with $\alpha \in \mathrm{End}^0(A) = M$. Rosati is CM conjugation, so symmetry gives $\alpha \in M^+$. The actual Galois action on M is cyclotomic, hence $c(\lambda) = \lambda$. In particular the minimum polarization $\lambda_{\min}$ is defined over $\mathbb{R}_{\mathrm{alg}}$. This does not identify it with μ or assert that it is rational.

Let $Y = A/H$ be a principal quotient with pullback $\lambda_{\min}$. If its polarized class is fixed by c, the exact root-orbit equivalence in Theorem 4.1 gives $c(H) = vH$ for $v \in \mu_{42}$. Multiplication by v^{-1} induces a polarized group isomorphism $f : Y^c \longrightarrow Y$. Its conjugate cocycle is

$$f\, c(f) = 1, \qquad v^{-1}\overline{v}^{-1} = 1.$$

This supplies a descent datum for the finite quadratic extension $\overline{\mathbb{Q}}/\mathbb{R}_{\mathrm{alg}}$. Since Y is projective, finite locally free descent is effective [16, Lemma 39.25.3, tag 0CCJ]. The origin, group law and polarization morphism descend by full faithfulness for fpqc descent of morphisms [16, Section 35.35, tag 02VV]. Conversely, a real polarized model forces its class to be fixed by c.

The real-embedding counts in Theorem 6.4 give $1+2\cdot 2+4\cdot 6 = 29$ conjugation-fixed underlying classes and the same number of minimum polarized classes including A. There are 28 principal fixed classes, two of order R and 26 of order S; the obstructed minimum orbits have no fixed classes. The free central involution exchanges the two principal classes over each principalizable underlying variety and commutes with c. Thus the 28 real principal classes lie over 14 underlying classes. Together with A this gives 15 real underlying classes. On each of the other 14 conjugation-fixed underlying classes, c exchanges its two principal classes.

It remains to exclude even unpolarized real models for these last classes. Suppose $Y_0/\mathbb{R}$ were such a model and choose a complex isogeny $\phi : A_{\mathbb{C}} \longrightarrow Y_{0,\mathbb{C}}$. The rank-one M-module of rational homomorphisms gives $c(\phi) = \phi u$ for $u \in M^\times$. Commutativity of M then gives

$$c(\phi\alpha\phi^{-1}) = \phi u\overline{\alpha}u^{-1}\phi^{-1} = \phi\overline{\alpha}\phi^{-1}.$$

So conjugation on the full endomorphism algebra of any such real form must be CM conjugation. Choose a real polarization μ_0 on Y_0. For any geometric polarization $\eta = \mu_0\beta$, Rosati symmetry puts $\beta \in M^+$. Consequently $c(\eta) = \eta$, and descent of morphisms makes every geometric polarization morphism real. In particular each principal class would be fixed, contradicting their exchange. These 14 classes are therefore pseudoreal. The remaining 160 underlying classes are not conjugation-fixed and cannot admit real models. This proves the asserted counts and distinctions. □

Corollary 7.2 (Real forms and forgetting the minimum polarization)**.** *The family in Theorem 7.1 has exactly* 58 *origin-preserving real minimum polarized forms:* 56 *principal forms, comprising four of order R and* 52 *of order S, and two forms of the minimum-polarized source. There are exactly* 29 *underlying real forms. Forgetting the minimum polarization has two elements in every fiber. All forms admit models over $\mathbb{R}_{\mathrm{alg}}$, and each real polarized automorphism group is $\{1, -1\}$.*

Proof. Fix a real model and let τ be its semilinear descent involution. The preceding theorem forces conjugation on the full endomorphism algebra to be CM conjugation. Its action on the full geometric polarized automorphism group G is therefore inversion. This group is μ_6 for every principal quotient and μ_{42} for the source. Every alternative polarized descent datum is $a\tau$ with $a \in G$, and $(a\tau)^2 = 1$. An isomorphism h between $a\tau$ and $b\tau$ exists exactly when $b = h^2a$. Effective descent thus identifies the real forms with G/G^2, which has two classes represented by 1 and -1. Fixed automorphisms satisfy $h = h^{-1}$ and hence are ± 1. This argument applies over both $\overline{\mathbb{Q}}/\mathbb{R}_{\mathrm{alg}}$ and $\mathbb{C}/\mathbb{R}$; the full automorphism groups agree under algebraically closed base extension.

For the underlying variety with full order T, replace G by $T^\times$. The cocycle quotient is

$$\frac{\{u \in T^\times : u\bar{u} = 1\}}{\{h/\bar{h} : h \in T^\times\}}.$$

The numerator consists of roots of unity. Put $E_L = \mathcal{O}_{M^+}^\times$ and $U_S = S^\times \cap M^+$. The adopted identities $R^\times = \mu_6 E_L$ and $S^\times = \mu_6 U_S$ give denominator μ_3, so each of the 14 real quotient complex classes has two underlying real forms. For $T = \mathcal{O}_M$, root squares give μ_{21} in the denominator, while the unit $\gamma = z^6(1 - z^2)$, $z = \zeta_{42}$, has $\gamma/\bar{\gamma} = -z^{14} \notin \mu_{21}$. The denominator is therefore all μ_{42}, leaving just one underlying real form of A. This gives $2 \cdot 14 + 1 = 29$ underlying forms.

For a fixed geometric principal class, the forgetful map on twist sets is the identity $\mu_6/\mu_3 \longrightarrow \mu_6/\mu_3$. Each underlying quotient complex class has two principal classes, so each underlying real form supports two real principal classes. For the source the map is $\mu_{42}/\mu_{21} \longrightarrow \mu_{42}/\mu_{42}$, with two classes above its unique underlying real form. This proves the 58-to-29 distribution. The classification concerns polarization morphisms and origin-preserving isomorphisms, without translations, real torsors or higher-degree polarization counts. □

8 A common field of definition

Theorem 8.1 (Common field of definition and underlying descent obstruction)**.** *Retain $M = \mathbb{Q}(\zeta_{21})$ and $\mathcal{E} = M(\sqrt{(3 + \sqrt{21})/2})$. Let Y be any of the* 188 *underlying quotient classes A/H in Theorem 4.3. For every number field $M \subseteq F \subset \overline{\mathbb{Q}}$, the class Y has a model over F if and only if $\mathcal{E} \subseteq F$. In particular none of these* 188 *classes descends to its embedded field of moduli, all of which lie in M. This negative assertion does not include the rational source A.*

There is a single twist $A^{\mathrm{tw}}/\mathcal{E}$ carrying the minimum polarization and with constant π-torsion. Its 33880 *order-*27 *subgroup kernels and quotient maps are defined over $\mathcal{E}$. Hence $\mathcal{E}$ is the smallest common field of definition inside $\overline{\mathbb{Q}}$ for the* 189 *underlying classes and also for their* 2969 *minimum polarized classes. The common underlying field of moduli is M, strictly smaller.*

Proof. The source is rational with all endomorphisms defined over M. Writing $\lambda_{\min} = \mu\alpha$ using its rational restricted Jacobian polarization gives $\alpha \in M^+$, so $\lambda_{\min}$ is defined over M. On $D = A[\pi] \simeq \mathbb{F}_{729}$, actual Galois action over M is therefore a continuous scalar character

$$q : G_M \longrightarrow U_{28}.$$

Indeed it is $\mathcal{O}_M/\mathfrak{P}$-linear and preserves the polarization pairing, since G_M fixes μ_3. The normalization unit in Theorem 4.2 may here be chosen to be 1. Thus q modulo U_{14} is the adopted normalized action, and Theorem 6.1 gives

$$\ker(q \bmod U_{14}) = G_{\mathcal{E}}.$$

Suppose Y_0/F represents A/H, with $F \supseteq M$, and choose the resulting degree-27 geometric isogeny $\phi : A \longrightarrow Y_0$. The rank-one M-module of rational homomorphisms gives $\sigma(\phi) = \phi a_\sigma$, $a_\sigma \in M^\times$, for $\sigma \in G_F$. Since G_F fixes M, this is a character. The map ϕ is defined over a finite extension, so its character has finite image and therefore lies in μ_{42}. This finite-order argument, rather than an assumed integrality of a quasi-isogeny ratio, is what makes a_σ an automorphism. Taking kernels shows $\sigma(H) = a_\sigma^{-1}H$, so G_F fixes the root orbit of H. If some q_σ lay outside U_{14}, a scalar in $U_{28} \setminus U_{14}$ would preserve H. Such a scalar has order 4 or 28 and generates a field of degree 2 or 6 over $\mathbb{F}_3$. It cannot preserve a three-dimensional subspace. Consequently $G_F \subseteq G_{\mathcal{E}}$ and $\mathcal{E} \subseteq F$.

For sufficiency, the character q restricted to $G_{\mathcal{E}}$ takes values in U_{14}. Reduction identifies $\mu_{14} \subset \mu_{42}$ with U_{14}, since its kernel on μ_{42} is μ_3. Let $v_\sigma \in \mu_{14}$ be the unique lift of q_σ. The roots are defined over M, so v is a character and $v_\sigma^{-1}\sigma$ is a coherent descent datum. It factors through a finite Galois extension of $\mathcal{E}$. Projectivity and descent of morphisms make it effective as a polarized abelian variety $A^{\mathrm{tw}}/\mathcal{E}$ [16]. The twisted action on D is trivial. Every subgroup H and quotient map is consequently defined over $\mathcal{E}$ [8, Chap. IV, Lemma 2.1]. For isotropic H, the quotient polarization with pullback $\lambda_{\min}$ descends. For obstructed H, use the minimum form with pullback $3\lambda_{\min}$; these represent all obstructed minimum classes by the root-orbit description above. The quotient construction commutes with base change, and the uniquely determined polarization morphism descends. This supplies every claimed model over $\mathcal{E}$.

Finally any common field of definition for all underlying classes contains their common field of moduli M. The necessity just proved then forces it to contain $\mathcal{E}$. For all minimum polarized classes, their common field of moduli already is $\mathcal{E}$. The simultaneous construction proves minimality in both cases. No smallest field of definition for an individual class over fields not containing M is specified. □

The individual existence of CM models over $\mathcal{E}$ also follows from the classical descent theorem of Milne, with its correction and addendum [6, 7]. That theorem concerns the fully marked polarized quotient triple, whose field of moduli here is $\mathcal{E}$, rather than the coarse underlying field contained in M. Its additional purity hypothesis holds: the quotient root group μ_6 is a direct summand of μ_{42}. The polarization-independent shorthand in Milne's Remark 2 does not remove the polarization requirement on the isomorphisms defining this field. Our construction gives, in addition, one explicit twist with constant π-torsion and all quotient maps simultaneously over $\mathcal{E}$; the underlying descent obstruction is proved separately above.

Theorem 8.2 (A common real number field for the real forms)**.** *Let c be the chosen complex conjugation and put $\mathcal{E}_{\mathbb{R}} = \mathcal{E}^{\langle c \rangle} = \mathcal{E} \cap \mathbb{R}$. All* 58 *minimum polarized real forms and all* 29 *underlying real forms classified above have models over $\mathcal{E}_{\mathbb{R}}$ inducing those real forms under the selected embedding. This field has degree* 12 *and signature* $(6, 3)$*; it is not totally real.*

Among the 14 *underlying real quotient complex classes, exactly* 12 *have field of moduli* M^+*. For each of these, and for each of its real forms,* $\mathcal{E}_{\mathbb{R}}$ *is the smallest real algebraic field of definition inside* $\overline{\mathbb{Q}} \cap \mathbb{R}$*. It is also the smallest common real algebraic field of definition for all* 15 *underlying real-model complex classes, for all* 29 *underlying real forms, and for all* 58 *minimum polarized real forms.*

Proof. Use the character q and its unique lift $v : G_{\mathcal{E}} \longrightarrow \mu_{14}$ from Theorem 8.1. On D, conjugation is semilinear for $x \mapsto x^{27}$, so

$$q(c\sigma c) = q(\sigma)^{27} = q(\sigma)^{-1}, \qquad v(c\sigma c) = c(v(\sigma)).$$

The last equality follows from injectivity of root reduction on μ_{14}. The sequence from $G_{\mathcal{E}}$ to $G_{\mathcal{E}_{\mathbb{R}}}$ splits by the actual involution c. Consequently $v(\sigma)^{-1}\sigma$ on $G_{\mathcal{E}}$ together with c gives a coherent finite descent datum on the source. The minimum polarization is defined over $M^+ \subset \mathcal{E}_{\mathbb{R}}$ and is preserved by these roots, so the datum is effective as a polarized abelian variety. Its D-action is trivial over $\mathcal{E}$ and is the original conjugation action over $\mathcal{E}_{\mathbb{R}}$. Only stability of D is needed; the endomorphism π itself is not asserted to be defined over $\mathcal{E}_{\mathbb{R}}$.

For a conjugation-fixed principal quotient class choose H and $r \in \mu_{42}$ with $c(H) = rH$. Replace the source datum at c by $r^{-1}c$. Its square is 1, and its cross-relation with $G_{\mathcal{E}}$ is unchanged because r is defined over $\mathcal{E}$ and commutes with the lifted roots. It preserves H and the source polarization. Finite subgroup quotient descent and uniqueness of the quotient polarization therefore give the desired principally polarized model over $\mathcal{E}_{\mathbb{R}}$. For the source itself use the preceding datum.

Twisting each model by -1 along $\mathcal{E}/\mathcal{E}_{\mathbb{R}}$ changes its real datum from τ to $-\tau$: under the selected real embedding this quadratic extension becomes $\mathbb{C}/\mathbb{R}$. The root-square classification above identifies these as exactly the two polarized real forms. This supplies all 58 forms, hence all 29 underlying forms by the proved forgetful surjection.

The four polarized size-12 orbits have 24 conjugation-fixed classes by signature $(6, 3)$. The two conjugation-fixed principal classes above each real underlying class therefore give 12 real underlying classes with field of moduli M^+. The other two real quotient classes, one of each order, have field of moduli $\mathbb{Q}(\sqrt{21})$. If an algebraic field $F \subset \overline{\mathbb{Q}} \cap \mathbb{R}$ defines one of the former 12 classes, it contains M^+. The model descends to a number field $F_0 \subseteq F$, which we enlarge inside F to contain M^+. Base change to F_0M and Theorem 8.1 imply $\mathcal{E} \subseteq F_0M \subseteq FM$. Since M/M^+ is imaginary quadratic, $(FM)^c = F$, whence $\mathcal{E}_{\mathbb{R}} \subseteq F$. The constructed models prove the individual and common minimality assertions. No minimality among arbitrary nonreal fields or transcendental real extensions is asserted. □

9 Minimum degrees and the remaining cases

Theorem 9.1 (Minimum degrees of underlying fields of definition)**.** *Let* Y *be one of the* 188 *underlying quotient classes, with embedded field of moduli* $k_Y \subseteq M$*. Every number field* F *defining* Y *satisfies* $2 \mid [F : k_Y]$*. For the* 144 *classes with* $k_Y = M$*, the minimum absolute degree of a field of definition is* 24*, attained by* $\mathcal{E}$*. For the* 24 *classes with* $k_Y = M^+$*, it is* 12*, attained by suitable conjugates of* $\mathcal{E}_{\mathbb{R}}$*. The latter assertion includes the* 12 *pseudoreal classes in those orbits, using nonreal embedded conjugates.*

For the remaining 16 *classes with field of moduli* B*, the lower bound is* 8*; for the remaining four with field of moduli* $\mathbb{Q}(\sqrt{21})$*, it is* 4*. No equality is asserted in these last two cases. The rational source* A *has minimum degree* 1*.*

Proof. A model over F forces $k_Y \subseteq F$. Its base change to FM and Theorem 8.1 give $\mathcal{E} \subseteq FM$, so $[FM : M]$ is even. Since $M/\mathbb{Q}$ is Galois, restriction identifies $\mathrm{Gal}(FM/F)$ with $\mathrm{Gal}(M/(M \cap F))$. The degree formula gives

$$[FM : M] = [F : M \cap F], \qquad [F : k_Y] = [F : M \cap F]\,[M \cap F : k_Y].$$

Thus $[F : k_Y]$ is even. The fields-of-moduli table gives the lower bounds and the counts $144 + 24 + 16 + 4 = 188$.

The common $\mathcal{E}$ models attain degree 24 when $k_Y = M$. Each of the four underlying size-6 orbits contains three real-model classes: its polarized size-12 orbit has six conjugation-fixed classes, with two above each real underlying class. Theorem 8.2 supplies an $\mathcal{E}_{\mathbb{R}}$ model for such a real member. Conjugating this model by an automorphism of $\overline{\mathbb{Q}}/\mathbb{Q}$ supplies a degree-12 model for every member of the same orbit. For a pseudoreal member the resulting embedded field cannot lie in $\mathbb{R}$, although its abstract signature is still $(6, 3)$. This proves degree minimality without asserting a unique smallest field under inclusion among arbitrary nonreal fields. □

10 Weil structures and maps from obstructed quotients

Corollary 10.1 (Weil structures and Hodge classes on the quotient family)**.** *All* 160 *principal polarized classes of Theorem 4.1, every minimum polarization on each of the* 108 *obstructed varieties of Theorem 5.1, and the minimum class on the actual Fermat factor A have integral $\mathcal{O}_K$ action for $K = \mathbb{Q}(\sqrt{-3})$, signature $(3, 3)$, and Hermitian discriminant class -1.*

Each of the 188 *underlying quotients, and A, has rational Hodge algebra generated by divisors and its two-dimensional K-Weil space in codimension three, with Hodge-dimension polynomial $(1 + t)^6 + 2t^3$. Every rational Hodge class on every finite product of these varieties, with repetitions allowed, is algebraic.*

Proof. Choose the minimum form E on A as the pullback of an actual principal B form through a degree-27 M-linear isogeny of Proposition 3.1. The accepted B form has signature $(3, 3)$ and discriminant -1 by Corollary I.8.7. Pullback gives a rational K-linear isometry, so E has the same discriminant. All 160 principal classes admit a quotient presentation with pullback E by Theorem 4.1; the required source unit normalization commutes with K. Their rational Hermitian spaces are therefore isometric to B's.

The full integral orders O, R, S all contain $\mathcal{O}_K = \mathbb{Z}[z^{14}]$, since $z^{14} \equiv 1 \pmod{\mathfrak{P}}$. The actual analytic K action retains signature $(3, 3)$ under the specified isogenies, and every polarization Rosati involution restricts to conjugation. For a minimum form $F = 3E_\varepsilon$ on an obstructed variety, Theorem 5.1 gives $\varepsilon \in E_L^+$. In common rational coordinates its Hermitian form is $h_F(x, y) = h_E(3\varepsilon x, y)$. Its determinant ratio is

$$N_{M/K}(3\varepsilon) = N_{L/\mathbb{Q}}(3\varepsilon) = 3^6 = 729 = N_{K/\mathbb{Q}}(27),$$

because $M = LK$, $L \cap K = \mathbb{Q}$, and $N(\varepsilon) = 1$. The discriminant class remains -1. The minimum forms on A differ by positive real units and have determinant ratio 1. None of these rational determinant statements changes the nonprincipal polarization types or the integral lattices.

Every quotient is M-linearly isogenous to A and hence B. These algebraic isogenies identify rational cohomology rings, divisor subspaces, and the rational Weil spaces $W_K(Y) = \bigwedge_K^6 H^1(Y, \mathbb{Q})$. Thus the B-type divisor/Weil generation and the displayed dimension polynomial transfer from the all-powers argument of Section H.2. The principal B model and the original trace-polarized B fiber

are isogenous, as established in the principal-lattice construction; their integral polarizations are not being identified.

A product X of n chosen factors is isogenous to B^n. Taking rational quasi-inverses in $\mathrm{Hom}\otimes\mathbb{Q}$ and clearing denominators gives an actual isogeny $f : B^n \to X$. Theorem H.2.3 supplies rational algebraicity on B^n. If a rational algebraic cycle Z represents $f^*\alpha$, then $(\deg f)^{-1} f_* Z$ represents α, since $f_* f^* = \deg f$ on rational cohomology. This proves algebraicity in every codimension for the stated products, all in one isogeny class. □

This corollary supplies rational algebraicity and ambient Weil structures. It gives no additional McMullen curve points, identification with Mostaed's literal normalization, integral Hodge statement, or explicit cycle equations.

Theorem 10.2 (Minimum principal-target isogenies from obstructed quotients)**.** *For each of the* 108 *obstructed varieties $Y = A/H$, the minimum degree of an isogeny to any principally polarized target is* 729*. The minimum degree of $Y \to A$ is* 27*; all such minimum maps have one kernel $C_0 \simeq (\mathbb{Z}/3)^3$ and differ by target automorphisms.*

For any fixed minimum polarization λ on Y, the maps to principal targets with pullback λ have exactly 89,360,320 *distinct kernels. Their group types and counts are*

$(\mathbb{Z}/3)^6$	91,840
$\mathbb{Z}/9 \oplus (\mathbb{Z}/3)^4$	89,268,480.

Exactly 1120 *contain C_0, equivalently factor through a minimum map $Y \to A$; the remaining* 89,359,200 *do not.*

Proof. Use the common lattices O and Λ for A and Y, with $[\Lambda : O] = 27$ and $\Lambda O = \mathfrak{P}^{-1}$. Every rational map is scalar in M. The condition $c\Lambda \subset O$ is equivalent, by taking O-saturations, to $c \in \mathfrak{P}$. Conversely $\mathfrak{P}\Lambda \subset O$, so

$$\mathrm{Hom}(Y, A) = \mathfrak{P} = \pi O, \qquad \deg(\pi u : Y \to A) = 27|N_{M/\mathbb{Q}}(u)|.$$

The minimum is 27, attained precisely for $u \in O^\times$. These maps differ by target units and have kernel $C_0 = \mathfrak{P}^{-1}/\Lambda$, of order 27 and exponent 3. This is the kernel of $\pi : Y \to A$, not of $\pi : Y \to Y$.

A degree-d map to a principal target pulls back a polarization of degree d^2. Theorem 5.1 gives $d \geq 729$. Composing $\pi : Y \to A$ with a degree-27 principal quotient of A attains this bound. More precisely, write the fixed minimum form on Y as $F = 3E_\varepsilon$, with $\varepsilon \in E_L^+$. Since $\pi\bar{\pi} = 3$, the map π pulls E_ε back to F. Thus the bound is attained with each prescribed minimum form as pullback.

Let $D = \Lambda^\#/\Lambda$ be the discriminant group for F. Its elementary divisors give the symplectic module

$$D \simeq (\mathbb{Z}/9)^2 \oplus (\mathbb{Z}/3)^8,$$

with one hyperbolic pair of denominator 9 and four of denominator 3. Integral self-dual overlattices correspond precisely to isotropic subgroups $J = J^\perp$ of order 3^6, by the dual-lattice argument of Theorem 4.1. These are exactly the kernels with fixed pullback F, including all possible principal targets.

The subgroup $D[3]$ has dimension 10 over $\mathbb{F}_3$, with pairing radical $3D$ of dimension 2. An elementary Lagrangian of dimension 6 must contain this radical: its image in the symplectic quotient $D[3]/3D$ has dimension at most 4. Such Lagrangians are therefore the inverse images of the Lagrangians in that eight-dimensional quotient, giving

$$L_4(3) = (3+1)(9+1)(27+1)(81+1) = 91840.$$

For a cyclic subgroup $C = \langle x \rangle$ of order 9, nondegeneracy supplies y with pairing $1/9$. The pair x, y generates a nondegenerate $(\mathbb{Z}/9)^2$ summand V, so $D = V \oplus W$ orthogonally. Since $3V$ already has order $9 = |3D|$, we have $3W = 0$; hence W is symplectic $\mathbb{F}_3^8$. Moreover $C^\perp = C \oplus W$. The Lagrangians containing C are exactly $C \oplus L$, with L a Lagrangian in W, so there are 91840 and each has type $\mathbb{Z}/9 \oplus (\mathbb{Z}/3)^4$. Every non-elementary Lagrangian contains an order-9 element, so this exhausts the remaining types.

There are

$$\frac{3^{12} - 3^{10}}{9 - 3} = 78732$$

cyclic subgroups of order 9 in D. Each non-elementary Lagrangian contains $(3^6 - 3^5)/(9 - 3) = 81$ such subgroups. Counting pairs (C, J) with $C \subset J$ therefore gives

$$\frac{78732 \cdot 91840}{81} = 89268480$$

non-elementary kernels. Adding the elementary population gives the stated total.

Finally, a fixed-form principal quotient factors through $r = \pi : Y \to A$ exactly when its kernel contains C_0. The resulting map from A has degree 27 and pullback E_ε, since pullback by r is injective on rational forms. Conversely every such principal quotient of A gives a kernel above C_0. The subgroup correspondence $J \mapsto r^{-1}(J)$ is bijective, so Theorem 4.1 gives exactly 1120. All minimum maps $Y \to A$ differ by target automorphisms, hence give the same factorization condition. □

The 89,360,320 count fixes a minimum pullback form. It counts subgroups, not polarized target classes, and is not multiplied by 26 when the pullback form varies: intersections between those kernel populations require a separate analysis.

The finite family consisting of A and its 188 underlying quotients has the following complete minimum-polarization data. Here $c_{\min}$ counts minimum polarization classes on one underlying variety, and d_{pp} is the minimum isogeny degree to a principal target.

Full order	$\#Y$	$\deg \lambda_{\min}$	Type	$c_{\min}$	d_{pp}
O	1	729	$(1,1,1,3,3,3)$	1	27
R	2	1	$(1,1,1,1,1,1)$	2	1
S	78	1	$(1,1,1,1,1,1)$	2	1
S	108	531441	$(1,3,3,3,3,9)$	26	729

The entries follow from Proposition 3.2 and Theorems 4.3, 5.1 and 10.2. Thus the 189 underlying varieties support exactly $1 + 2 \cdot 2 + 78 \cdot 2 + 108 \cdot 26 = 2969$ minimum polarized classes. Different underlying varieties cannot merge as polarized varieties. Corollary 10.1 gives their integral Weil structures and rational Hodge algebraicity on finite products. This is a finite Fermat three-torsion quotient family, distinct from the corrected $\ell = 43$ Hecke packet and from a curve-point census.

11 A structural two-prime comparison

Proposition 11.1 (The two-prime primitive factor)**.** *Let $p < q$ be distinct odd primes, $m = pq$, and let H_d denote the smooth projective curve $y^2 = x^d - 1$. In $J(H_m)$ let A_m be the connected image of the primitive Φ_m projector, with its actual inclusion. Its dimension is $(p-1)(q-1)/2$, and the canonical restricted polarization has type*

$$(1^{[(p-2)(q-1)/2]},\ p^{[(q-p)/2]},\ (pq)^{[(p-1)/2]}),$$

where bracketed superscripts denote multiplicities. Both this polarization and the addition isogeny $A_m \times J(H_p) \times J(H_q) \to J(H_m)$*, using pullback along* $(x,y) \mapsto (x^q, y)$ *and* $(x,y) \mapsto (x^p, y)$*, have degree* $q^{p-1}p^{q-1}$*. For the covariant rotation* T *of* $J(H_m)$*, put*

$$\mathcal{N}_m = pq - p\sum_{j=0}^{q-1} T^{pj} - q\sum_{j=0}^{p-1} T^{qj}.$$

Then $\operatorname{im}\mathcal{N}_m = A_m$*,* $\mathcal{N}_m$ *is Rosati symmetric, and* $\mathcal{N}_m^2 = pq\mathcal{N}_m$*. The least positive denominator of the projector* $e_m = \mathcal{N}_m/(pq)$ *in* $\operatorname{End}(J(H_m))$ *is* pq*.*

Proof. The prime-degree ramification argument of Proposition 2.2 applies to both actual pullbacks: they are embeddings with induced forms $q\lambda_{H_p}$ and $p\lambda_{H_q}$. Their integral separation holds uniformly. Indeed choose $1 \le a < p$ with $qa = 1 + kp$. The identity

$$\left(\sum_{j=0}^{a-1} t^{qj}\right)\Phi_q(t) - t\left(\sum_{l=0}^{k-1} t^{pl}\right)\Phi_p(t) = 1$$

follows by expanding the two consecutive geometric sums. Each cyclotomic polynomial annihilates the corresponding lower Jacobian image, so those images have zero intersection, including torsion. Their character spaces are orthogonal. The lower product's polarization kernel is consequently

$$(\mathbb{Z}/q)^{p-1} \oplus (\mathbb{Z}/p)^{q-1} \simeq (\mathbb{Z}/p)^{q-p} \oplus (\mathbb{Z}/pq)^{p-1}.$$

The holomorphic differentials $x^{i-1}dx/y$ and their conjugates give every nontrivial character of the order-m rotation group once in full cohomology. The remaining characters have order pq, so the actual A_m is the connected orthogonal complement of the lower product. The saturated unimodular lattice argument of Proposition 2.2 transfers the displayed discriminant group to A_m. Its nonunit elementary divisors give $(q-1)/2$ entries; the dimension supplies the stated number of unit entries. The common group order gives both claimed degrees.

The two deck sums act by q and p on their respective lower character spaces and vanish on the primitive part. Thus e_m is the orthogonal primitive projector. Inversion preserves the deck sums, proving Rosati symmetry. Its polynomial residual is $\mathcal{N}_m^2 - pq\mathcal{N}_m = 2pq(1 + T + \cdots + T^{pq-1})$; the full trace factors through the Jacobian of the genus-zero rotation quotient and therefore vanishes. Finally, the discriminant group has exponent pq. The lattice-projection proof of Corollary 2.3 therefore proves that pq is the least integral denominator. □

This uniform statement concerns the induced polarization and the specified projector. It requires neither simplicity of A_m nor a classification of its other polarizations.

12 The restricted principal census and Jacobian question

This section supplies the order-and-type result invoked for the source factor. Its arithmetic proof is independent of the quotient and descent sections. The theta step has an additional execution and enclosure contract; the exact conditional statement is retained below.

Theorem 12.1 (Complete simple principal classification with the selected order)**.** *There are exactly ten origin-preserving polarized isomorphism classes of simple principally polarized complex abelian sixfolds* X *such that* $\operatorname{End}^0(X) \simeq M$ *and, under this identification,* $R \subseteq \operatorname{End}(X)$*. The containment condition is independent of the field identification. Eight classes have endomorphism ring* R *and*

polarized automorphism group μ_6; two have endomorphism ring $\mathcal{O}_M$ and polarized automorphism group μ_{42}. They belong to four unpolarized isogeny classes, with the following counts at one CM-type representative from each orbit:

Φ	*flip parity from* Φ_0	$[R]$	$[\mathfrak{a}]$	$[\mathcal{O}_M]$
$\{1,5,11,13,17,19\}$	0	2	2	0
$\{1,5,11,13,19,25\}$	1	0	0	1
$\{1,5,11,17,23,29\}$	0	2	2	0
$\{1,5,11,23,25,29\}$	1	0	0	1

In particular, the four specified polarized B/C objects exhaust the objects in their unpolarized isogeny class satisfying this R-containment condition. No assertion is made here for other endomorphism orders or for the induced A type.

Proof. Every automorphism of M preserves $\mathcal{O}_L$ and takes $1-z^{14}$ to itself or an associate. Hence it preserves $R = \mathcal{O}_L + (1-z^{14})\mathcal{O}_M$. After choosing a field identification, $H_1(X,\mathbb{Q})$ has dimension one over M, and its integral lattice is a full fractional R-ideal. Conversely, the positive unimodular forms of Proposition I.7.11 give abelian varieties by the Riemann criterion.

We check full endomorphisms and simplicity, not merely a centralizer. For a CM type Φ, put $H = \{\sigma \in \mathrm{Gal}(M/\mathbb{Q}) : \sigma\Phi = \Phi\}$. The splitting of $\mathrm{End}_{\mathbb{Q}}(M) = \bigoplus_\sigma M\sigma$ over $\mathbb{C}$ has disjoint permutation-matrix supports for distinct σ. A nonzero coefficient in M has nonzero value at every embedding, so a term crossing the two Hodge blocks cannot cancel. Thus the full rational Hodge endomorphism algebra is $M \rtimes H = \mathrm{End}_{M^H}(M)$. The rational weight-one equivalence [9, Corollary 6.9] and Poincaré complete reducibility show that the abelian variety is simple precisely when H is trivial: otherwise this matrix algebra has nontrivial idempotents, whereas for trivial H it is the field M.

The exact census of the 64 types gives precisely the four trivial-stabilizer orbits in the table, each of size 12. The two retained implementations are `experiments/fixed_field_cm_census_v98/primary.py` and `experiments/fixed_field_cm_census_v98/independent.py`. The bounded table derivation `experiments/simple_cm_unmarked_v145/derive.py` binds their accepted outputs, checks orbit coverage and parity, and applies Proposition I.7.11; it does not re-prove the ideal classification or the unit-norm theorem.

Every polarization on a simple object here has Rosati involution equal to conjugation on M. Indeed its positive trace form extends to $M \otimes \mathbb{R} \simeq \mathbb{C}^6$. An involution exchanging factors would give trace norm zero on a factor idempotent. On a fixed complex factor, the identity would give negative trace norm on i. Positivity therefore forces conjugation on every factor. Thus the marked classification covers every principal polarization.

Fix one primitive type representative. An unmarked polarized isomorphism conjugates the full endomorphism fields and hence is semilinear for some $\sigma \in \mathrm{Gal}(M/\mathbb{Q})$. Holomorphicity at this fixed type forces $\sigma\Phi = \Phi$, so $\sigma = 1$. Forgetting the marking therefore identifies no additional classes or automorphisms at the chosen representative. Every object in its orbit can be remarked to this representative, because R is Galois-stable. The table and automorphism assertions now follow from Proposition I.7.11.

An unpolarized isogeny likewise induces a field automorphism relating the types, so distinct primitive orbits are not isogenous. Within one orbit, after remarking, full fractional lattices in M are commensurable; an integer multiple of the identity gives an isogeny. Finally $\sigma_{31}(\Phi_C) = \Phi_B$ places the four distinct objects of Proposition I.7.12 in the first orbit. Its count of four proves the restricted exhaustion assertion. □

Corollary 12.2 (Jacobian membership in the ten-class list)**.** *Of the ten classes in Theorem 12.1, the four in the third row and the single class in the second row are not canonically polarized Jacobians of smooth curves. The single class in the fourth row is the known Jacobian of the smooth projective model of $y^7 = x^3 - 1$. These signature and automorphism criteria alone leave the four B/C classes in the first row undecided; Theorem 12.4 will exclude them using the invariant theta divisor. If any is a Jacobian, its curve is nonhyperelliptic, has full automorphism group C_3, and is a cyclic triple cover of $\mathbb{P}^1$ with eight branch points, four of each nonzero inertia exponent.*

Proof. Strong Torelli [12, Theorem 12.1] realizes an odd-order polarized Jacobian automorphism β by a curve automorphism up to sign. Squaring the latter induces β^2 and has the same odd order, by faithfulness on the Jacobian. For $\beta = \zeta_3$, there are no invariant differentials, so the curve quotient has genus zero. Riemann–Hurwitz gives eight fixed points. If n_1, n_2 count the two inertia exponents, then $n_1 + n_2 = 8$ and $n_1 + 2n_2 \equiv 0 \pmod 3$. The cyclic-cover eigenspace formula [4, equation (2.2)] gives only signatures $(4,2), (3,3), (2,4)$, respectively for $n_2 = 1, 4, 7$. The third row has signature $(1,5)$ and is therefore excluded. This is also directly Zarhin's established criterion $g + 2 < 3|a - b|$ [17, Theorem 1.1], here $8 < 12$; the obstruction method is not new. The first row has signature $(3,3)$ and hence requires $n_1 = n_2 = 4$. A hyperelliptic curve would induce a nontrivial order-three map on its hyperelliptic base $\mathbb{P}^1$, with only two fixed points and at most four curve points above them. This contradicts eight fixed points. Strong Torelli and the full polarized automorphism group μ_6 then give full curve automorphism group C_3.

For either maximal-order object, use instead $\beta = \zeta_{21}$. Its eigencharacters are primitive, so again the curve quotient has genus zero. If a, b, c count branch points with ramification indices $3, 7, 21$, Riemann–Hurwitz gives $7a + 9b + 10c = 26$. Its only nonnegative solution is $(1,1,1)$. After fixing these three branch points at $0, 1, \infty$, the monodromy residues are $(7u, 3v, -7u - 3v)$, with $u \in (\mathbb{Z}/3)^\times$ and $v \in (\mathbb{Z}/7)^\times$. The Chinese remainder theorem makes these twelve tuples a single orbit under a change of cyclic generator. Kummer theory thus gives one curve up to complex isomorphism, namely $y^7 = x^3 - 1$. Its known simple maximal-order CM type is $\{1, 2, 4, 8, 10, 16\}$ modulo 21 [4, Proposition 6.11, Lemma 6.12 and Proposition 6.14]. Its odd lifts modulo 42 are $\{1, 23, 25, 29, 31, 37\}$, which are 23 times the fourth row's representative. That row is therefore the known Jacobian. The distinct second-row object cannot also be a Jacobian. The finite ramification and type arithmetic is retained in `experiments/simple_cm_unmarked_v145/jacobian_check.py`; the proof uses the geometric inputs just named and makes no assertion about isogeny to a Jacobian. □

12.1 An invariant theta obstruction and the Jacobian classification

Write B_0, B_1, C_0, C_1 for the four principal objects of Proposition I.7.12, where the subscript records the prescribed polarization or its $4 - t^2$ twist. Throughout this subsection, Jacobian means the canonical principal polarization of a smooth curve, up to origin-preserving polarized isomorphism.

Proposition 12.3 (The balanced triple-cover count)**.** *Let C be a smooth genus-six cyclic triple cover of $\mathbb{P}^1$ with eight branch points, four of each nonzero inertia exponent. If δ is a deck generator and $G = \ker(1 - \delta) \subset J(C)$, the unique δ-invariant symmetric theta divisor has exactly 213 points of G outside its support. Its multiplicities on G are $0, 1, 2$ with respective frequencies $213, 456, 60$.*

Proof. Move infinity away from the branch locus and write $y^3 = \prod_{i=1}^{8}(x - t_i)^{a_i}$, with $a = (1,1,1,1,2,2,2,2)$. Let P_i be the ramification points and H the fiber at infinity. Then $3P_i \sim H$ and $D = \sum_i P_i - H$ has degree five and $2D \sim K_C$. It is the unique invariant theta characteristic: the characteristic torsor has translation group $J(C)[2]$, on which $1 - \delta$ is invertible since $(1 - \delta)(2 + \delta) = 3$.

Put $W = \{b \in \mathbb{F}_3^8 : \sum_i b_i = 0\}$. For representatives $b_i \in \{0,1,2\}$, define

$$E_b = \sum_i b_i P_i - \frac{\sum_i b_i}{3} H.$$

The map $b \mapsto [E_b]$ takes values in G and has kernel $\langle a \rangle$. Indeed a gives $\mathrm{div}(y)$; conversely, a function with invariant principal divisor is a δ-eigenfunction $y^j h(x)$, and its orders at the P_i force $b_i = ja_i$ modulo three. Both $W/\langle a \rangle$ and G have order 3^6, the latter by the eigenvalues of $1-\delta$ on $H^1(C,\mathbb{C})$. This identifies every point of G.

For $m = 1 + \sum_i b_i/3$, the three eigensheaves for sections of $\mathcal{O}_C(D+E_b)$ on $\mathbb{P}^1$ have degrees

$$e_k = -m - 4k + \sum_i \left\lfloor \frac{ka_i + 1 + b_i}{3} \right\rfloor, \qquad k = 0,1,2.$$

This follows by taking orders of $y^k h(x)$ at the branch points and infinity. Thus $h^0(D+E_b) = \sum_k \max(e_k+1, 0)$, independently of the distinct branch positions. The exact finite count on the 729 classes gives sorted degree triples

$(e_0,e_1,e_2)_{\mathrm{sorted}}$	$(-1,-1,-1)$	$(-2,-1,0)$	$(-2,-2,1)$	$(-3,0,0)$	$(-3,-1,1)$
frequency	213	456	24	24	12.

The retained finite certificate is specified in `experiments/bc_jacobian_v147/adoption.json`; its domain is exactly the quotient just proved complete. Riemann's singularity theorem identifies these dimensions with theta multiplicities [2, Introduction, smooth-curve statement]. Translation by D identifies the theta support in $\mathrm{Pic}^0(C)$ with $h^0(D+E_b) > 0$. The principal line bundle has $h^0 = 1$, so its invariant class determines a unique effective divisor. The displayed frequencies give the assertion. □

Theorem 12.4 (Jacobian classification under the enclosure contract)**.** *Assume the trusted Arb/acb enclosure and execution contract specified below for the retained theta certificates. Exactly one of the ten classes in Theorem 12.1 is a canonically polarized Jacobian of a smooth curve: the known class of the smooth projective model of $y^7 = x^3 - 1$. In particular, none of B_0, B_1, C_0, C_1 is such a Jacobian.*

Proof. For each model, $\delta = \zeta_3$ preserves the principal polarization and satisfies $\delta^2 + \delta + 1 = 0$. Symmetric representatives form a torsor under two-torsion, so the same invertibility argument gives a unique invariant line-bundle class. A principal ample line bundle has a one-dimensional space of sections, hence determines a unique effective divisor in this class. Its support is stable under negation. If a B/C model were a Jacobian, the necessary geometry in Corollary 12.2 would identify this divisor with the one in Proposition 12.3. Inverting the deck generator does not change its fixed kernel or invariant divisor.

The retained certificates give, on each model separately, 107 distinct nonzero classes modulo sign in $\ker(1-\delta)$ at which the theta section is nonzero. The exact-coordinate sets are the indexed representatives $0, \ldots, 106$ in the retained pair map. Their counts are

model	B_0	B_1	C_0	C_1
nonzero pairs	107	107	107	107
distinct points	214	214	214	214.

The kernel is killed by three, so a nonzero point differs from its negative. Each model therefore supplies $214 > 213$ points outside the invariant divisor, contradicting the Jacobian count. No origin value or sum across different models is used. Combining these four exclusions with Corollary 12.2 proves the tally. □

The analytic computation starts from the exact ideal lattice, actual CM embeddings and principal alternating form. It verifies an integral symplectic basis, recomputes the δ action and its unique invariant quadratic refinement in that basis, and constructs interval periods. For normalized lattice $[I, \tau]$, a fixed point has the form $z = (n + \tau m)/3$ with $(1 - F)(n, m)^t = 0$ modulo three. The half-characteristic is read from that same refinement. These conventions and their geometric identification are reviewed in `experiments/bc_jacobian_v147/PERIOD_SETUP_REVIEW.md`.

The nonvanishing records are exact dyadic rectangles for squared theta values. An independent rational check verifies that a real or imaginary interval excludes zero and that all representatives are disjoint modulo sign. This check does not reimplement the analytic evaluator: the conclusion assumes correctness of the pinned Arb/acb routines, the stated enclosure/serialization interface, and the admitted execution and runtime provenance. An enclosure excluding zero for the square implies nonvanishing of the section; it does not determine a zero's multiplicity. The cumulative certificate is `experiments/bc_jacobian_v147/certificate_run07b_adoption.json`, which binds the eight accepted retained packages; the geometric consequence is bound by `experiments/bc_jacobian_v147/jacobian_classification_adoption.json`. The separate hostile review in `experiments/bc_jacobian_v147/session_v150/CLASSIFICATION_REVIEW03.md` checks all 428 raw rectangles and the complete implication above, including the two explicitly admitted runtime variants. It does not prove the analytic library implementation or its equivalence to source.

The theorem concerns the specified endomorphism-order category and polarized isomorphism. It does not exclude isogenies to Jacobians or address arbitrary CM orders, and establishes no additional Hodge algebraicity or Hilbert-modular orbit membership.

13 The smaller-order Jacobian cases

The following criterion describes what a Jacobian in the S-order family would have to satisfy. Partial nonvanishing computations do not settle its 156 polarized classes.

Proposition 13.1 (A structural theta filter for balanced cubic actions)**.** *Let (Y, λ) be a complex principally polarized abelian sixfold with a polarized automorphism δ satisfying $\delta^2 + \delta + 1 = 0$ and tangent signature $(3, 3)$. Its unique δ-invariant symmetric principal line-bundle class has a unique effective divisor Θ. Normalize the order-three bundle lift to fix the origin fiber. In $G = \ker(1 - \delta)$ there are 261 points with trivial fiber character and 234 with each nontrivial character. The theta section has trivial character; therefore the 468 points in the nontrivial sectors lie on Θ. Every zero in the 261-point sector is singular.*

Proof. Write $\Lambda = H_1(Y, \mathbb{Z})$, let E be its unimodular Riemann form, and let F act as δ. Since $(1 - F)(2 + F) = 3$, the affine action on the torsor of symmetric representatives has a unique fixed class. Its semicharacter is $(-1)^{q(k)}$ for the invariant quadratic refinement q of E modulo two. In the Appell–Humbert convention $H(x, y) = E(x, Iy) - iE(x, y)$, where I is the complex structure, the multiplier is $j_k(z) = (-1)^{q(k)} \exp(\pi H(z, k) + \pi H(k, k)/2)$; see [14, §1.2, pp. 7–8]. The lift $T(z, t) = (Fz, t)$ has order three and fixes the origin fiber.

Put $N = F - I_{12}$. Fixed points are $x = -(F + 2I_{12})k/3$, parametrized by $\Lambda/N\Lambda$. This quotient has order 3^6 and is killed by three, since $N^2 = -3F$ and $\det N = 3^6$. Set $b(k) = E(k, Fk)$. Invariance of q gives $q(k) = b(k) \pmod 2$, and $E(3x, k) = b(k)$. The normalized fiber character is consequently

$$j_k(x)^{-1} = \exp\big(\pi i(3q(k) + b(k))/3\big) = \zeta_3^{2b(k)}.$$

Modulo three, the polar form of b is $B_b(k, l) = -E(k, Nl)$. Also $b(Nu) = 0$ and $N^2 = 0$ modulo three, so b descends to $\Lambda/N\Lambda$. Unimodularity makes the polar radical on $\Lambda/3\Lambda$ equal to $\ker N =$

$\operatorname{im} N$, both of dimension six. Thus the descended quadratic form is nondegenerate.

Its six-variable Gauss sum is $+27$ or -27: diagonalization expresses it as a product of six sums $\pm i\sqrt{3}$. The inverse lift $\delta^* L \to L$ has the conjugate fiber sum, which has the same real value. Holomorphic Lefschetz [1, Theorem 4.12] applies, with simple fixed points and complex denominator $(1-\zeta_3)^3(1-\zeta_3^2)^3 = 27$. Kodaira vanishing and Riemann–Roch give $H^{>0}(Y, L) = 0$ and $h^0(Y, L) = 1$. The section eigenvalue is therefore $+1$ or -1; its order divides three, so it must be $+1$. The Gauss sum is 27. Character orthogonality gives $261, 234, 234$ for the three levels of b. Equivariance forces the nontrivial sectors to vanish. At a zero in the trivial sector, the first derivative intertwines the tangent action with the trivial fiber action. There is no invariant tangent covector, so this derivative vanishes as well. □

Corollary 13.2 (Necessary Jacobian geometry of the smaller-order targets)**.** *If any of the* 156 *S-order principal classes of Theorem 4.1 is a canonical smooth-curve Jacobian, its curve is non-hyperelliptic, is a cyclic triple cover of $\mathbb{P}^1$ branched at eight points with inertia multiplicities four and four, and has full automorphism group C_3. For the invariant symmetric theta divisor, the fixed-point multiplicity distribution is*

fiber sector	0	1	2
trivial (261)	213	0	48
nontrivial (468)	0	456	12.

The origin is outside the divisor. Among the 130 *eligible nonzero opposite pairs there are then* 106 *nonvanishing pairs and* 24 *double-zero pairs.*

Proof. Corollary 10.1 supplies the integral cubic action and signature, and Theorem 4.1 gives the full polarized automorphism group μ_6. The signed Torelli, Riemann–Hurwitz, cyclic-cover eigenspace and hyperelliptic arguments of Corollary 12.2 therefore apply with the same hypotheses, giving the asserted curve geometry and full automorphism group. Proposition 12.3 gives total multiplicities $213, 456, 60$. Proposition 13.1 puts all nonvanishing points in the trivial sector and makes its zeros singular. This gives the displayed table. At the origin, the three eigensheaf degrees in the proof of Proposition 12.3 are all -1, so its invariant theta characteristic has no section. Negation now gives the pair counts. □

The proof and regression are retained in `experiments/s_order_jacobian_v160/`. In particular, the split quadratic-form sign follows structurally from the balanced tangent action; it is not extrapolated from the four B/C matrices. No theta evaluation on these 156 smaller-order classes is asserted, and their Jacobian status remains open. A certificate of nonvanishing on 107 eligible nonzero opposite pairs would exclude the particular principal object tested; the B/C certificates do not supply such values on any new target.

The classical two-prime isogeny decomposition appears in [3, Eq. (4.2) and Lem. 4.2]. The following argument computes the restriction on the specified embedded primitive factor.

14 Questions left by the classification

Three finite questions remain particularly concrete.

1. Which of the 156 full-S principal classes are canonical Jacobians? The ten-class R-containing theorem does not answer this question. A partial theta count is evidence only until the required threshold and all model, enclosure and execution hypotheses have been met.

2. Are the definition-degree lower bounds 8 and 4 sharp for the remaining 16 and 4 underlying classes, respectively? A minimum degree is different from a unique smallest embedded field of definition.

3. For each of the 108 obstructed varieties, what is the union of principal-target kernels over its 26 minimum polarization forms? The fixed-form count cannot simply be multiplied by 26, since kernels may recur and target classes may coincide.

These questions extend the proved classification and do not enter its proofs. The construction supplies a finite testing ground for integral CM geometry and descent; it asserts neither new points on McMullen's curve nor a general classification of abelian sixfolds.

References

[1] M. F. Atiyah and R. Bott, A Lefschetz fixed point formula for elliptic complexes: II. Applications, *Annals of Mathematics* (2) **88** (1968), no. 3, 451–491; Theorem 4.12. https://webhomes.maths.ed.ac.uk/~v1ranick/papers/abott2.pdf.

[2] S. Casalaina-Martin and J. L. Kass, *A Riemann singularity theorem for integral curves*, Amer. J. Math. **134** (2012), no. 5, 1143–1165. Introduction, classical smooth-curve statement; arXiv:0907.0212v2. https://arxiv.org/abs/0907.0212v2

[3] H. Goodson, *An exploration of degeneracy in abelian varieties of Fermat type*, arXiv:2211.03909v2, 1 August 2024. The cited numbering refers to this version. https://arxiv.org/abs/2211.03909v2

[4] W. Li, E. Mantovan and R. Pries, *Data for Shimura varieties intersecting the Torelli locus*, arXiv:2105.02286v1 (2021); Pacific J. Math. **343** (2026), 179–210. Equation (2.2), Proposition 6.11, Lemma 6.12 and Proposition 6.14 refer to the inspected arXiv version. https://arxiv.org/abs/2105.02286v1

[5] J. M. Masley, Solution of small class number problems for cyclotomic fields, *Compositio Mathematica* **33** (1976), 179–186, https://www.numdam.org/item/CM_1976__33_2_179_0.pdf.

[6] J. S. Milne, Abelian varieties defined over their fields of moduli, I, *Bull. London Math. Soc.* **4** (1972), 370–372. https://www.jmilne.org/math/articles/1972c.pdf

[7] J. S. Milne, Correction to "Abelian varieties defined over their fields of moduli, I", *Bull. London Math. Soc.* **6** (1974), 145–146, including the addendum.

[8] J. S. Milne, *Abelian Varieties*, course notes, version 2.0, March 2008, Chapter IV, Lemma 2.1, p. 135 (quotients by Galois-stable finite subgroups). https://www.jmilne.org/math/CourseNotes/AVc.pdf

[9] J. S. Milne, *Introduction to Shimura Varieties*, revised September 16, 2017, Theorem 6.8 and Corollary 6.9, https://www.jmilne.org/math/xnotes/svi.pdf.

[10] J. S. Milne, *Class Field Theory*, version 4.03, 6 August 2020. https://www.jmilne.org/math/CourseNotes/CFT.pdf

[11] J. S. Milne, *Complex Multiplication*, version 0.10, 14 July 2020, Section 3.11 and Propositions 3.12–3.13 and 3.17, pp. 29–31. https://www.jmilne.org/math/CourseNotes/CM.pdf

[12] J. S. Milne, *Jacobian Varieties*, revised June12, 2021, Theorem 12.1, p. 38; original chapter in *Arithmetic Geometry* (Springer, 1986), 167–212. https://www.jmilne.org/math/xnotes/JVs.pdf

[13] OpenAI math repository, *The rational Hodge conjecture for CM abelian varieties*, public manuscript dated 6 October 2026, introduction (main CM theorem) and tensor-reduction section (auxiliary-factor contractions). The comparison uses repository commit `4eaee87548250b2682affda7d05d947c05554bae`; the general theorem and its formalization are not proof inputs here. https://github.com/openai/math/tree/4eaee87548250b2682affda7d05d947c05554bae/preprints/The-rational-Hodge-conjecture-for-CM-abelian-varieties-October-6-2026

[14] A. Polishchuk, *Abelian Varieties, Theta Functions and the Fourier Transform*, Cambridge University Press, section 1.2, pp. 7–8. Publisher excerpt: https://assets.cambridge.org/97805218/08040/excerpt/9780521808040_excerpt.pdf.

[15] K. Saller, *Split Weil structures on abelian sixfolds and the Hodge conjecture*, version 2, Zenodo record 23290280 (2026), Remarks 4.7–4.8. https://doi.org/10.5281/zenodo.23290280

[16] The Stacks Project Authors, *The Stacks Project*, Lemma 39.25.3 (tag 0CCJ), and Section 35.35 (tag 02VV), finite locally free effective descent and fpqc descent of morphisms. https://stacks.math.columbia.edu/tag/0CCJ; https://stacks.math.columbia.edu/tag/02VV.

[17] Y. G. Zarhin, *Cubic surfaces and cubic threefolds, jacobians and intermediate jacobians*, arXiv:math/0610138v3 (2007), Theorem 1.1. https://math.nyu.edu/~tschinke/.manin/final/zarhin/zarhin.pdf